Geospatial Prior Guided Confused-Class Data Augmentation for Semantic Segmentation of Remote Sensing Images

Jingying Li, *Graduate Student Member, IEEE*, Guoying Liu, Jiaxuan Li, Tiancheng Ma, Leiguang Wang, and Chen Zheng, *Member, IEEE*

***Abstract*—Semantic segmentation of high spatial resolution remote sensing images plays a crucial role in diverse applications. Deep learning-based methods have emerged as the dominant approach in this field. However, existing methods face the challenge of effectively using the information in annotated training data, resulting in limited model performance. Data augmentation is a commonly used technique to better reveal the information embedded in the data through appropriate transformations. However, these methods usually fail to account for the unique characteristics of remote sensing images, such as the high spectral similarity among different classes and intraclass variation among distinct objects. This oversight hinders the model's ability to distinguish between spectrally similar classes, resulting in persistent misclassifications. To address these challenges, we propose a novel geospatial prior guided confused-class data augmentation (GCA) method for semantic segmentation of remote sensing images. By emphasizing the occurrence of confused classes within augmented training data, GCA compels the model to focus on learning discriminative features to improve class separability. GCA consists of three key modules: the geospatial prior information extraction module (module *G*), the confused-class similar units detection module (module *C*), and the confused-class data augmentation module (module *A*). Specifically, module *G* extracts geographic units and their spatial relationships. Module *C* identifies highly similar units from confused classes that are prone to misclassification. Module *A* generates challenging training data by emphasizing confused-class similar units under the guidance of geospatial priors. Experiments on the Gaofen Image Dataset (GID) and LoveDA datasets demonstrate that the proposed GCA method shows a better performance than state-of-the-art data augmentation methods. The GCA code is publicly available at https://github.com/Jingying-Li/GCA**

Received 7 July 2025; revised 1 December 2025 and 7 February 2026; accepted 3 March 2026. Date of publication 9 March 2026; date of current version 16 March 2026. This work was supported in part by the Natural Science Foundation of Henan under Grant 232300421071, in part by the National Natural Science Foundation of China under Grant 32160369 and Grant 41961053, in part by the Program for Innovative Research Team (in Science and Technology) in University of Henan Province under Grant 26IRTSTHN003, and in part by the State Key Laboratory of Spatial Datum. *(Corresponding authors: Leiguang Wang; Chen Zheng.)*

Jingying Li, Jiaxuan Li, Tiancheng Ma, and Chen Zheng are with the School of Mathematics and Statistics and Henan Engineering Research Center for Artificial Intelligence Theory and Algorithms, Institute of Applied Mathematics, Henan University, Kaifeng 475004, China (e-mail: 104752210083@henu.edu.cn; lijiaxuan_student@163.com; 15226036576@163.com; zhengchen_data@126.com).

Guoying Liu is with Ministry of Education, Key Laboratory OBI Information Processing, Anyang 455000, China (e-mail: lgydatabase@163.com).

Leiguang Wang is with the College of Landscape Architecture and Horticulture, Southwest Forestry University, Kunming 650224, China (e-mail: wlgbain@126.com).

Digital Object Identifier 10.1109/TGRS.2026.3671763

***Index Terms*—Confused classes, data augmentation, deep learning, geospatial prior information, semantic segmentation.**

I. Introduction

SEMANTIC segmentation is a key task in high spatial resolution remote sensing image analysis. It aims to classify each pixel into a specific land cover or land use class, and has been widely used in agricultural management [1], environmental monitoring [2], disaster assessment [3], and urban planning [4]. In recent years, deep neural networks (DNNs) have attracted increasing attention in semantic segmentation due to their outstanding performance [5], [6], [7]. The evolution of DNNs, from early end-to-end architectures to more advanced techniques such as attention mechanisms and transformers, has effectively improved segmentation accuracy in remote sensing applications.

Although DNNs have achieved remarkable success in remote sensing, their performance is highly dependent on the quantity and quality of training data [8], [9], [10]. However, obtaining large-scale datasets with pixel-level annotations is challenging due to the labor-intensive and costly manual labeling process [11], [12], [13], [14]. At the same time, the performance of DNNs depends not only on the amount of annotated training data but also on the effective utilization of information contained within the existing annotated training data. In fact, DNNs usually fail to capture the generalization and discriminative features adequately. For instance, DNNs are highly sensitive to subtle distribution shifts and often suffer performance degradation even with minor differences between training and test data [15]. As shown in Fig. 1, when randomly selected training data from raw data are used as test data, the DNN trained on raw data yields segmentation results that are closer to the ground truth. However, when subtle transformations are applied to the selected training data to generate test data, the segmentation performance degrades significantly, with such transformations including rotation, horizontal slicing, and vertical slicing. These results suggest that DNNs fail to effectively use the information contained in the training data. Hence, the full exploitation of existing annotated training data remains a challenging problem.

To alleviate this challenge, many advanced methods have been proposed. They can be roughly categorized into two groups: generative-based methods and data augmentation-based methods. The former aims to generate synthetic training

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.

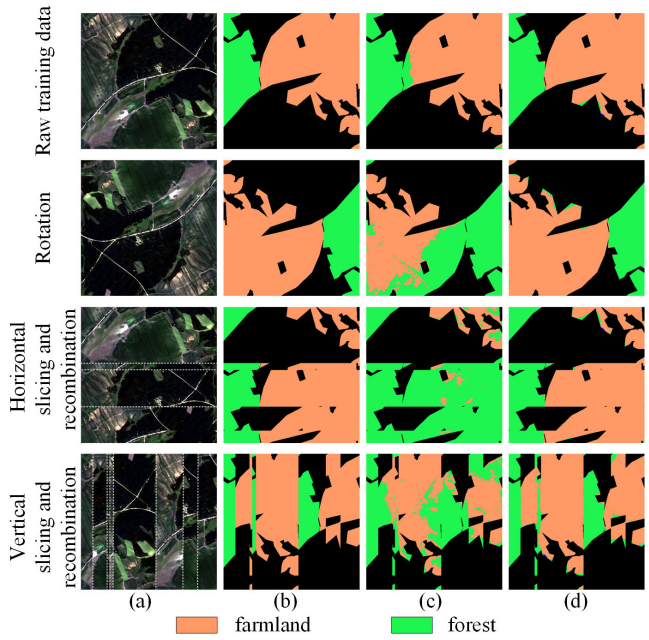


Fig. 1. Illustration of training data utilization differences between DNNs trained on raw data and GCA, based on segmentation results from raw training data and its subtly transformed variants. (a) Raw training data and its subtly transformed variants, (b) ground truth, (c) results from the DNN trained on the raw data, and (d) results from the DNN trained with GCA.

data by leveraging neural networks to learn the underlying data distribution, such as generative adversarial networks (GANs) and diffusion models [16], [17]. Specifically, GANs generate diverse synthetic training data through adversarial training between a generator and a discriminator. However, GANs commonly suffer from training instability and mode collapse, which negatively affect the quality of the generated data [18], [19]. In contrast, diffusion models iteratively denoise random noise to generate high-quality images, effectively facilitating model convergence and mitigating the mode collapse issue [20]. Nevertheless, these methods typically incur high computational costs [21] and may introduce misleading image details that deviate from ground truth due to noise [22], [23]. The latter, data augmentation-based methods, aim to highlight crucial generalization features that DNNs struggle to learn through appropriate transformations, thereby enabling better utilization of existing annotated training data and promoting the learning of discriminative features. They can be divided into three main types: geometric augmentation, color augmentation, and spatial context-based augmentation. First, geometric augmentation uses geometric transformations to strengthen the robust representation ability of DNNs in terms of object sizes, orientations, and viewpoints, such as cropping [24], [25], rotation [26], scaling [27], and flipping [28]. Second, color augmentation uses spectral transformations to improve robustness against variations in spectral and illumination, such as adjustments of brightness, contrast, and hue [29], [30], [31]. Finally, the spatial context-based augmentation method uses regional contextual transformations to enhance the model's ability to utilize contextual information and strengthen the learning of local discriminative features. Specifically, cutout introduces a region-based regularization strategy by randomly masking rectangular regions, encouraging the model to infer missing details from the surrounding context and focus on the full context of the image [32]. Building on cutout, CutMix [33] replaces the removed regions with patches from other training data, introducing informative local content while preserving essential global contextual information [34]. Furthermore, mixup [35] generates new samples by linearly interpolating the pixel values and labels of two training data, encouraging the model to learn smoother feature transitions and improving generalization.

Although data augmentation improves segmentation accuracy by enhancing the model's ability to utilize information, most existing approaches usually focus on common information such as geometric structures, color distributions, or spatial context. However, they still face challenges in addressing the characteristics of remote sensing images, such as geospatial information, interclass similarity, and intraclass variation among distinct objects. To tackle these challenges, many methods have been recently proposed to highlight the characteristics of remote sensing images. Specifically, positionaware graph-CNN fusion network (PGCFN) integrates positional information into the multiclass change detection process, improving the model's ability to capture spatial structural patterns [36]. Ship-Go uses diverse instances and environmental prompts to generate training data with varied backgrounds, enhancing the quality of background generation [37]. Category grouping-basic unit extraction-random combination (CBR) strategy exploits the intraclass spectral variability to enhance the model's intraclass generalization ability by randomly incorporating different intraclass units into augmented training data [38]. The above methods improve model performance by emphasizing intraclass generalization. However, remote sensing images contain not only intraclass spectral variability but also interclass spectral similarity, and most existing methods fail to account for interclass similarity. Moreover, due to interclass similarity, the last-layer features of different classes may converge toward similar class-mean directions in the representation space, which is characteristic of the neural collapse phenomenon in DNNs [39], leading to class confusion, as shown in Fig. 2(a). Meanwhile, the features learned by DNNs trained with CutMix and CBR still exhibit collapse patterns similar to those of the model trained on raw data, as illustrated in Fig. 2(b) and (c). This is because these methods focus only on intraclass spectral variability while neglecting interclass similarity. Therefore, effectively leveraging high spectral similarity among different geographic objects is essential for improving model performance.

To address these issues, this article proposes a novel geospatial prior guided confused-class data augmentation (GCA) method for semantic segmentation of remote sensing images. Guided by geospatial prior information, GCA augments easily confused classes by explicitly increasing the number of training data with confused-class similar units, compelling the model to learn more discriminative features and thereby reducing class confusion. GCA consists of three major modules, i.e., geospatial prior information extraction module (module G), confused-class similar units detection module (module C), and confused-class data augmentation module (module A). Module G is designed to extract geographic

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.

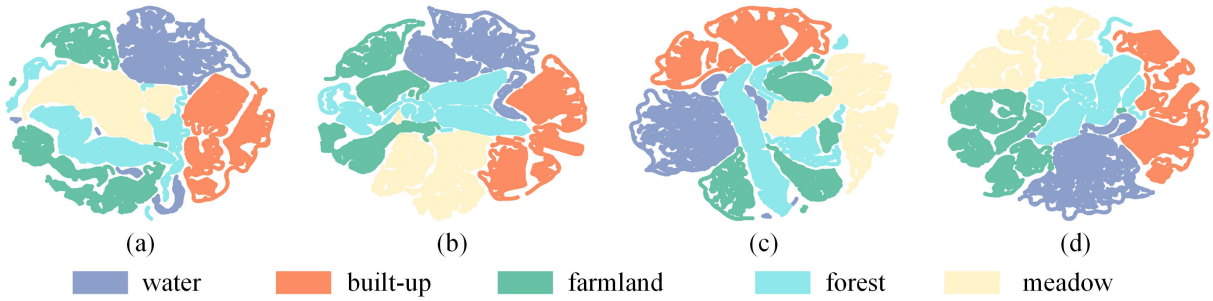


Fig. 2. Comparison of neural collapse patterns in DNNs trained with GCA and other data augmentation methods, visualized using t-SNE on last-layer features: (a) raw data, (b) CutMix, (c) CBR, and (d) GCA. Compared to the first three methods, the last-layer features learned by the DNN trained with GCA exhibit tighter intraclass distributions and clearer interclass separation, indicating that GCA encourages the model to learn more discriminative features by emphasizing confused classes.

units and their spatial relationships from the pixel-wise annotated images, serving as a geospatial prior and offering guidance based on geographic knowledge for the following data augmentation modules. Module C identifies highly similar geographic units belonging to easily confused classes, effectively capturing the challenging confused-class pairs and promoting the learning of discriminative features. Module A generates augmented training data that highlights confused classes to improve class separability, guided by geospatial prior information. By augmenting highly similar units from easily confusable classes, the last-layer features learned by the DNN trained with GCA exhibit more compact intraclass clusters and reduced overlap between different classes, as illustrated in Fig. 2(d). Moreover, the segmentation results shown in Fig. 1 indicate that the DNN trained with GCA yields more accurate segmentation, not only on selected training data but also on their subtle transformed variants. These results demonstrate the effectiveness of GCA in guiding the model to learn robust and discriminative features.

The main contributions of this article lie in the following three aspects.

1) A GCA method is proposed for remote sensing semantic segmentation. By increasing both the occurrence probability and the diversity of confused-class similar units, GCA encourages models to learn more discriminative features and improves segmentation performance.
2) Two augmented training datasets based on Gaofen Image Dataset (GID) and LoveDA are generated to highlight the unique characteristics of remote sensing images. These datasets explicitly emphasize hard-to-distinguish instances, resulting in enhanced confused-class representation and more effective model training.
3) Extensive experiments across multiple datasets using different DNNs demonstrate that the proposed GCA data augmentation method improves class separability compared with state-of-the-art data augmentation methods.

II. Proposed Method

The proposed GCA method aims to improve the separability of classes prone to misclassification by augmenting the training data with samples drawn from highly confusable classes. It contains three modules, i.e., geospatial prior information extraction module (G), confused-class similar units detection module (C), confused-class data augmentation module (A), as illustrated in Fig. 3. Module G extracts geographic units and their spatial relationships that serve as the geospatial prior and the basic augmentation unit for the subsequent modules. Based on these geographic units, module C identifies confused-class pairs and extracts their similar units. At last, module A strengthens these similar units and generates augmented training data under the guidance of the geospatial prior obtained from module G. The details of modules G, C, and A are discussed in the following.

A. Geospatial Prior Information Extraction Module

The geospatial prior information extraction module, module G, is designed to identify geographic units and capture their spatial relationship from the pixel-wise annotated images as the geospatial prior, as shown in Fig. 3. To identify geographic units, a superpixel-based method $\Gamma(\cdot)$ is applied to the annotated image L to obtain image partitions with clear semantic land classes. Specifically, let I denote the raw observed image and L denote its annotated image. The superpixel-based method $\Gamma(\cdot)$ is applied to the annotated image L to generate a set of superpixel regions, i.e.,

$$\Gamma(L) = \{L_s \mid s \in S\} \tag{1}$$

where each L_s represents a superpixel region indexed by s, and S is the index set of all superpixel regions. The mean shift algorithm is employed as the superpixel method $\Gamma(\cdot)$, with the minimum region area set to 0.05% of the total number of pixels in each training data and all other parameters kept at their default values. Since the superpixel method is performed on the annotated image L, each superpixel region L_s contains only one land class, as illustrated in Fig. 4(i). For each superpixel region L_s, we define R_s as the index set of pixels belonging to L_s, as shown in Fig. 4(ii). Given that L and I are spatially aligned and share the same spatial resolution, R_s can be directly applied as a region mask to I to extract the corresponding geographic unit G_s without any additional transformations, as illustrated in Fig 4(iii). Consequently, geographic units G_s with adaptively determined scales driven by local spatial and semantic homogeneity are extracted. Geographic units G_s extracted in this manner not only preserve the spectral–spatial characteristics of their land

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.

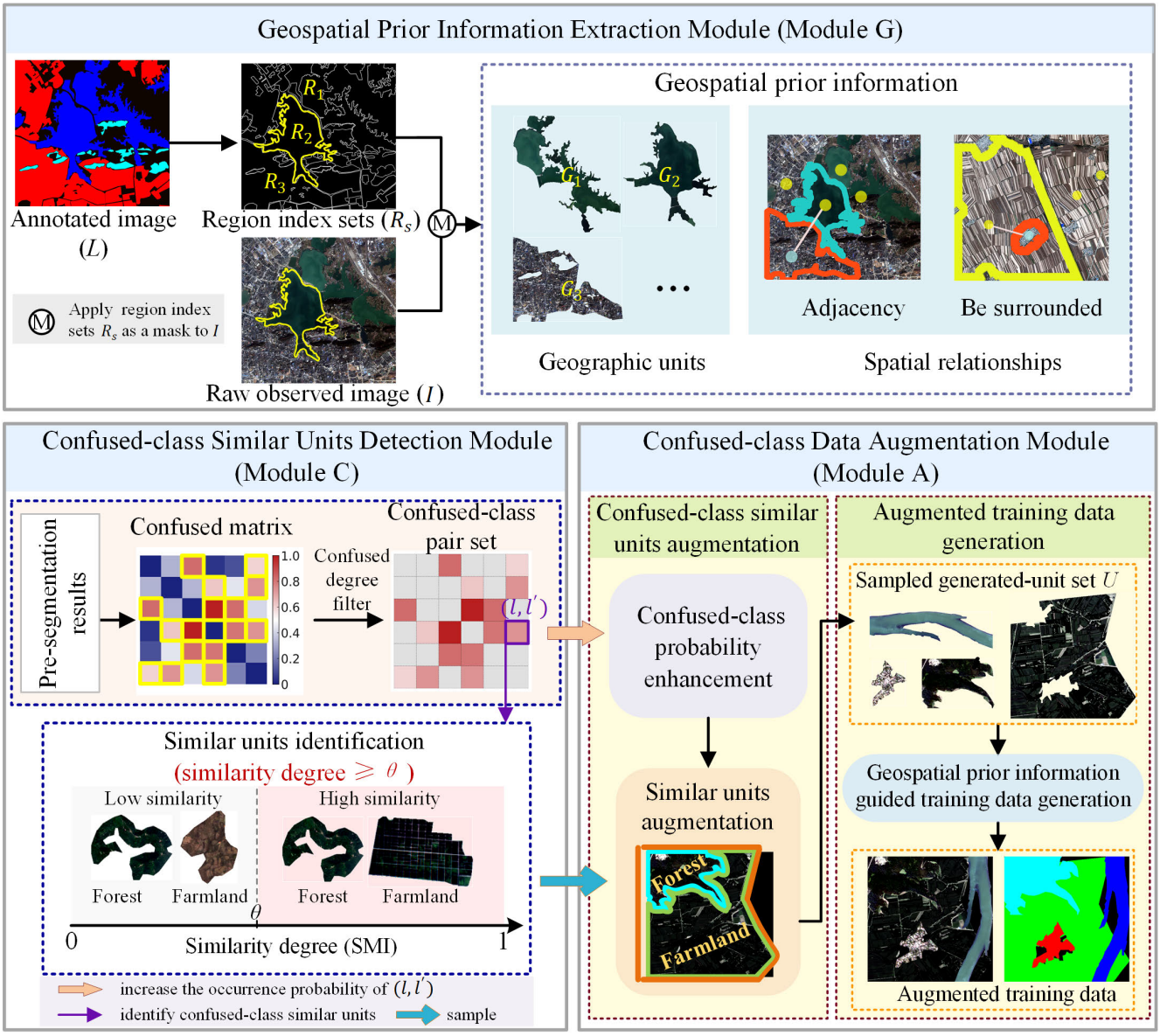


Fig. 3. Flowchart of the proposed GCA method, which consists of three modules: geospatial prior information extraction (module G), confused-class similar units detection (module C), and confused-class data augmentation (module A). Module G extracts geographic units and their spatial relationships. Module C detects highly similar geographic units from easily misclassified classes. Module A generates augmented training data by emphasizing confused-class similar units under the guidance of geospatial prior information.

classes, but also retain the inherent intraclass diversity arising from different spatial regions. Therefore, the geographic units G_s can be used as basic elements for constructing the augmented training data.

Based on the above identified geographic units, geospatial relationships between different land classes can be further captured in module G, i.e., adjacency and be surrounded. These two types of geospatial relationship are quantified into two matrices, denoted as $D = \{d_{ll'}\}_{K\times K}$ and $B = \{b_{ll'}\}_{K\times K}$. Here, l and l' denote land classes that take values from $\Lambda = \{1, \ldots, K\}$ and K is the number of land classes. For each $d_{ll'}$ in D, it is defined as

$$d_{ll'} = \sum_{s\in S_l} d_s\left(l'\right) \tag{2}$$

with

$$d_s\left(l'\right) = \begin{cases} 1, & \text{if } l' \in N_s \text{ and } |N_s| \geq 2 \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where $S_l = \{s \in S \mid \text{class}(G_s) = l\}$, $N_s = \{l' \in \Lambda \mid l' \text{ is the land class of geographic unit that neighbors } G_s\}$, $|N_s|$ is the number of elements in N_s, and $|N_s| \geq 2$ means that there are at least two land-cover classes are adjacent to G_s. For each $b_{ll'}$ in B, it is defined as

$$b_{ll'} = \sum_{s\in S_l} b_s\left(l'\right) \tag{4}$$

with

$$b_s\left(l'\right) = \begin{cases} 1, & \text{if } l' \in N_s \text{ and } |N_s| = 1 \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Here, $|N_s| = 1$ means that the current geographic unit G_s is adjacent only to land class l', i.e., it is surrounded by l'.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.

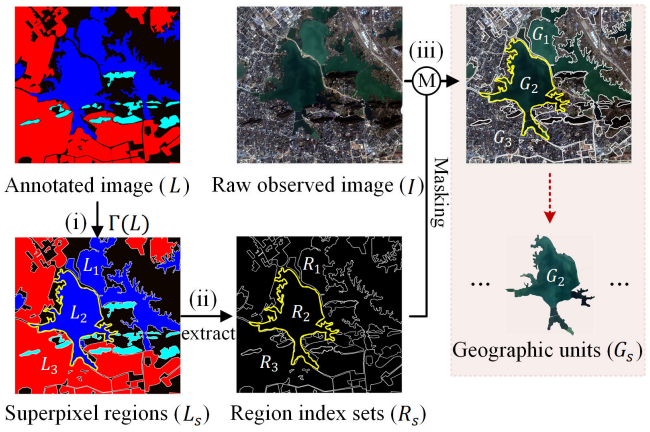


Fig. 4. Illustration of extracting geographic units by masking the raw observed image with region index sets derived from the annotated image.

In module G, geographic units and their two types of spatial relationships are collected as a geospatial prior. It is possible to capture the characteristics of each land class and the pattern of geospatial distribution between different land classes, thereby providing geospatial knowledge guidance for subsequent data augmentation modules. The geospatial priors in Module G are derived from the existing annotated data and reflect the spatial patterns observed in the training set, and are inherently data-dependent.

B. Confused-Class Similar Units Detection Module

The module C is developed to detect similar geographic units from confused classes that are prone to misclassification, as shown in Fig. 3. These units will be used to enhance confused-class separability. A two-stage strategy is proposed in this module. The first stage is to recognize confused classes by using the confusion matrix. Specifically, the confusion matrix is computed from DNN-based presegmentation results, denoted as $M = [M_{ll'}]_{K\times K}$. Here, $M_{ll'}$ represents the probability of class l being misclassified as l'. Then, the confused degree $C(l,l')$ between classes l and l' is computed as $C(l,l') = M_{ll'} + M_{l'l}$. The average value of $C(l,l')$ is used as the threshold to define the set of confused classes $\mathcal{C}$, i.e.,

$$\mathcal{C} = \left\{ (l,l') \,\middle|\, C(l,l') \geq \frac{\sum_{l,l'} C(l,l')}{K(K-1)} \right\}. \tag{6}$$

For clarity, each pair (l,l') in $\mathcal{C}$ is called a confused-class pair in the following. The second stage is to detect geographic units that show high spectral similarity between each confused-class pair (l,l'). Let $H_{G_s} \in \mathbb{R}^{N\times 1}$ denote the histogram vector of unit R_s, where $s \in S_l$, and N represents the dimension of the histogram vector. Its normalized form is computed as

$$H_{G_s}^{\text{nor}} = \frac{H_{G_s}}{\left\| H_{G_s} \right\|_1}. \tag{7}$$

The similarity degree between units G_s and $G_{s'}$ is then defined as

$$\text{SMI}(G_s, G_{s'}) = \sqrt{\left(H_{G_s}^{\text{nor}}\right)^{\top} H_{G_{s'}}^{\text{nor}}}. \tag{8}$$

Given a predefined similarity threshold θ, if $\text{SMI}(G_s, G_{s'}) \geq \theta$, G_s and $G_{s'}$ are identified as confused-class similar units, and forming the similar-unit set $(l,l')_{\mathcal{C}}$.

C. Confused-Class Data Augmentation Module

The module A is designed to highlight the confused-class similar units when generating the training data, aiming to improve the model's discriminative capability. This module comprises two main components: confused-class similar units augmentation and augmented training data generation. Specifically, the first component selects geographic units from different classes, where the occurrence probability of confused-class similar units is emphasized during the sampling process. The second component integrates the sampled geographic units into one training data, guided by the geospatial prior information to ensure spatial relationship coherence.

1) Confused-Class Similar Units Augmentation: This section aims to increase the occurrence probability of confused-class similar units during the geographic unit selection process. Specifically, the process consists of confused-class probability enhancement and sampling corresponding similar units to augment these pairs.

To achieve this, the annotated images L are first clipped into M raw data patches. For the ith patch, its class combination is denoted as $\tilde{\Lambda}_i$, and all extracted combinations constitute the set $\tilde{\Lambda} = \{\tilde{\Lambda}_i \mid 1 \leq i \leq M\}$. Because different raw data patches may have the same class combination, the unique class-combination set Λ of size N_Λ is obtained by removing duplicates from $\tilde{\Lambda}$, and its mathematical definition is $\Lambda = \{\Lambda_i \mid 1 \leq i \leq N_\Lambda\}$. The probability of each class-combination Λ_i is then defined as $P(\Lambda_i)$

$$P(\Lambda_i) = \frac{1}{M} \sum_{j=1}^{M} \mathbb{I}_{\{\tilde{\Lambda}_j = \Lambda_i\}}, \quad i \in \{1, \ldots, N_\Lambda\} \tag{9}$$

where $\mathbb{I}_{\{\tilde{\Lambda}_j=\Lambda_i\}}$ is the indicator function. To emphasize the occurrence probability of confused-class pairs, an enhanced weight w_i is assigned to adjust the probability of each $P(\Lambda_i)$, i.e.,

$$P'(\Lambda_i) = \frac{P(\Lambda_i) \times w_i}{\sum_i P(\Lambda_i) \times w_i} \tag{10}$$

where

$$w_i = \begin{cases} \sum_{(l,l')\in\mathcal{C}} \exp\left\{C(l,l')\right\}, & \text{if any } (l,l') \in \mathcal{C} \text{and } l,l' \in \Lambda_i \\ 1, & \text{otherwise.} \end{cases} \tag{11}$$

In above (10), as w_i containing confused-class pair (l,l') is larger than that without, the class combination with (l,l') will have greater chance to be sampled from the enhanced probability $P'(\Lambda_i)$, as shown in Fig. 5.

Second, for the specific class-combination Λ_i sampled from $P'(\Lambda_i)$, geographic units of corresponding classes will be directly collected as a generated-unit set U. Especially, if (l,l') is included in Λ_i, the confused classes l and l' only collect units from their similar units set $(l,l')_{\mathcal{C}}$. In other

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.

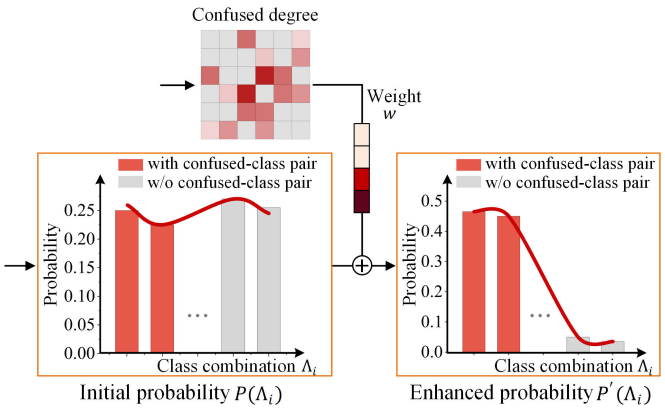


Fig. 5. Illustration of confused-class probability enhancement.

words, by augmenting similar units of confused-class pairs, the proposed method can focus on the hard-to-distinguish instances, thereby enhancing the separability of the confused classes. Notably, this unit collection process primarily augments confused-class similar units to alleviate interclass confusion, while also sampling diverse units from the same class to promote intraclass variability. By randomly sampling units with diverse spectral–spatial appearances, the generated-unit set U represents the spectral–spatial heterogeneity within each class, enabling the subsequent generation of training data to effectively capture intraclass diversity.

2) Augmented Training Data Generation: This section aims to generate training data T by integrating the augmented geographic units of U under the guidance of spatial relationships of the module G, as shown in Fig. 6. During the generation process, the spatial relationships of adjacency and be surrounded are used as the prior information to ensure consistency between the generated augmented training data and the real-world spatial distribution.

The first step uses the spatial relationship of be surrounded to update the geographic units in U. Specifically, for a unit of class l, if this class l may be surrounded by other class l' with probability $(b_{ll'})/(d_{ll'} + b_{ll'})$ based on (4), a random number $\tau_1 \sim \text{Uniform}(0, 1)$ will be generated. If $\tau_1 \le (b_{ll'})/(d_{ll'} + b_{ll'})$, this unit will be embedded into the unit of the class l' and the embedded units will be updated and treated as one unit in the second step, as illustrated in Fig. 6(b). If $\tau_1 > (b_{ll'})/(d_{ll'} + b_{ll'})$, this unit will be directly used in the second step.

The second step employs the spatial relationship of adjacency to integrate the updated units as the generated training data T. Namely, the updated units are first sorted according to their sparsity [38] and size. Each unit is then sequentially placed to generate T. During this placement process, for the current unit with class l, the probability $P_{ll'}$ that class l is adjacent to another class l' is calculated to determine potential adjacent classes, which is defined as $P_{ll'} = (d_{ll'})/\left(\sum_{l' \in \Lambda_i} d_{ll'}\right)$ based on (2). Subsequently, a random number $\tau_2 \sim \text{Uniform}(0, 1)$ is generated to select the adjacent class l' based on the cumulative probability distribution of $P_{ll'}$, i.e.,

$$l' = \begin{cases} l_1, & \text{if } 0 < \tau_2 \le P_{ll_1} \\ l_k, & \text{if } \sum_{j=1}^{k-1} P_{ll_j} < \tau_2 \le \sum_{j=1}^{k} P_{ll_j},\ k \in \{2, \ldots, |\Lambda_i|\} \end{cases}. \tag{12}$$

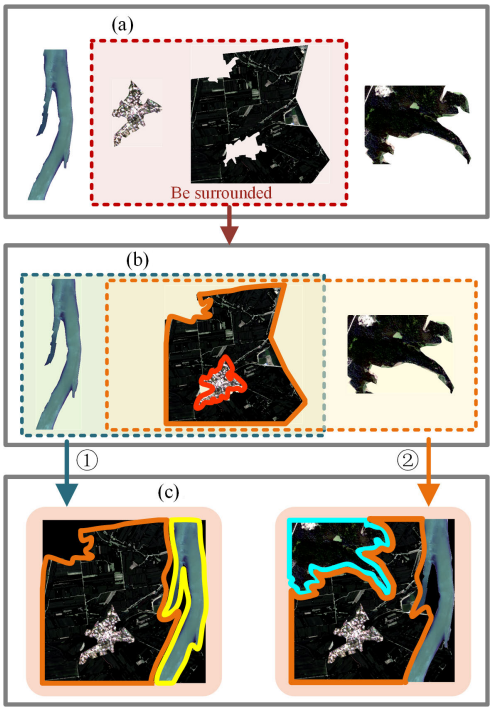


Fig. 6. Illustration of the augmented training data generation process guided by geospatial prior information, which leverages two types of spatial relationships: adjacency and be surrounded. (a) Sampled generated-unit set U. (b) Be surrounded by embedding. (c) Adjacency placing.

After determining l', the unit of class l is spatially placed adjacent to units of class l', as shown in Fig. 6(c). This placement process is repeated until all sorted units are integrated to generate a new training data T.

III. Experiments

The proposed GCA method generates augmented training data containing confused-class similar units, guiding the model to focus on highly confusable features and improve class separability. To comprehensively evaluate the effectiveness and robustness of GCA, different DNN-based methods with GCA are evaluated over various types of remote sensing images and compared with other state-of-the-art data augmentation methods. This section includes five modules. First, datasets and implementation details are provided. Second, the augmented datasets generated by GCA are introduced on three semantic segmentation benchmarks. Third, GCA is compared with other state-of-the-art data augmentation methods on the aforementioned datasets across ten different DNNs. Fourth, the parameter setting for the proposed GCA is discussed. Fifth, a comparison of the additional time cost with other data augmentation methods is conducted. Finally, an ablation study is conducted to evaluate the contributions of key components of GCA.

A. Datasets and Implementation Details

1) *Datasets:*

a) *GID-5 dataset [40]:* The GID comprises high-resolution remote sensing images with a spatial resolution of 4 m, acquired by the Gaofen-2 (GF-2) satellite. It covers over 50 000 km^2 across more

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.

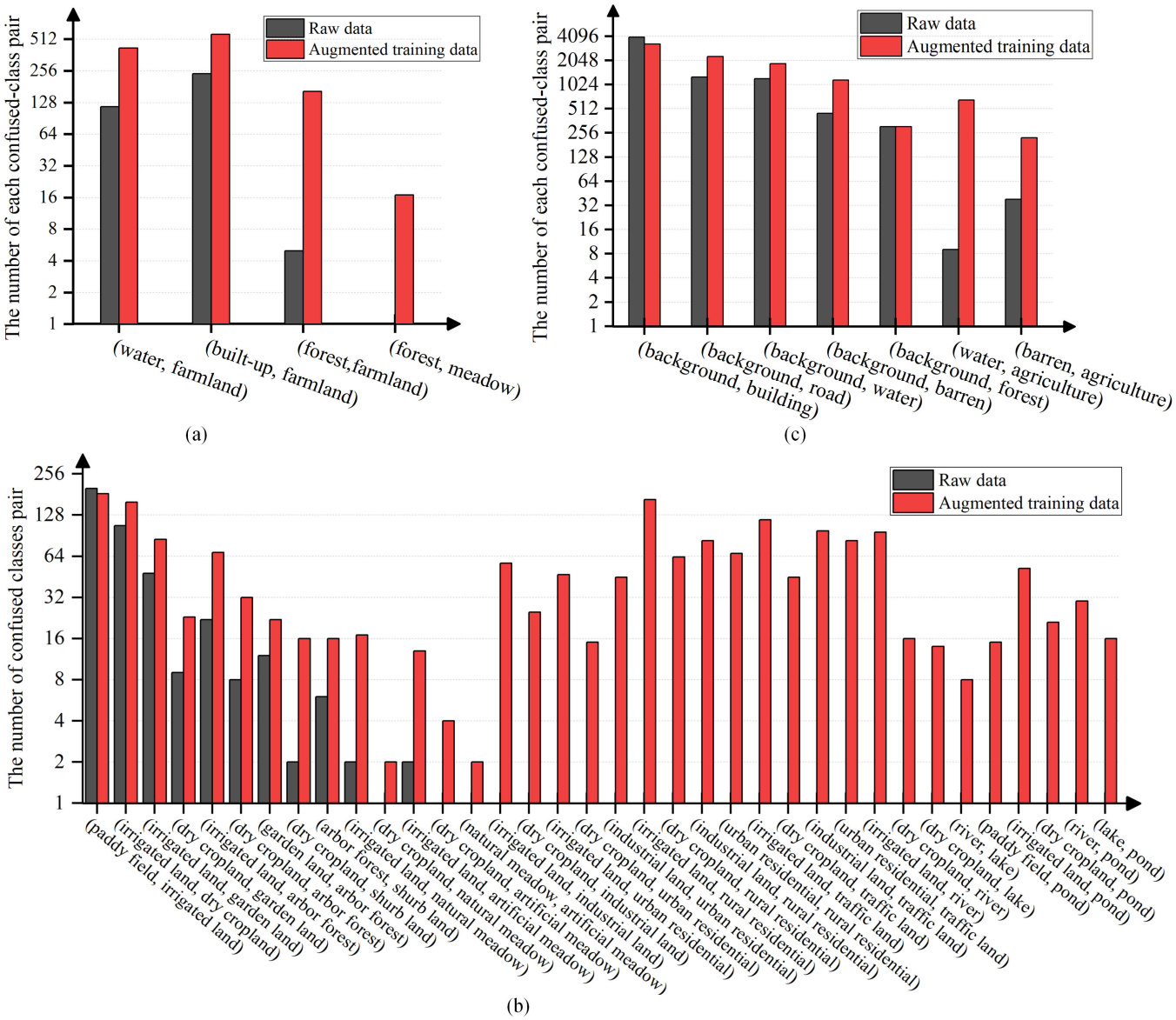


Fig. 7. Number of each confused-class pair in the raw data and the augmented training data across three semantic segmentation datasets. Note that the y-axis is plotted on a logarithmic scale to improve the visibility of low-frequency confused-class pairs. (a) GID-5 dataset. (b) GID-15 dataset. (c) LoveDA dataset.

than 60 cities. The GID-5 dataset consists of 150 GF-2 images, each with a size of 6800 × 7200 pixels. Five land-cover classes are annotated, including water, built-up, farmland, forest, and meadow. In this experiment, 120 annotated images are used for training, while the remaining 30 are used for testing. To ensure fairness, GCA and all compared augmentation methods use 3000 training data for training.

b) *GID-15 dataset [41]:* GID-15 is an extension of the GID-5 dataset, developed by refining the five land-cover classes into 15 subclasses for more fine-grained classification tasks. These subclasses include paddy field, irrigated land, dry cropland, garden land, arbor forest, shrub land, natural meadow, artificial meadow, industrial land, urban residential, rural residential, traffic land, river, lake, and pond. In this experiment, the same 120 images used in GID-5 are reannotated with 15-class labels for training, while ten additional annotated images are used for testing. The training data setting is the same as that of GID-5.

c) *LoveDA dataset [42]:* The LoveDA dataset provides a comprehensive set of remote sensing scenes, including urban and rural landscapes. Each image in the dataset has a spatial resolution of 0.3 m and a size of 1024 × 1024 pixels. The dataset encompasses seven land-cover classes: background, building, road, water, barren, forest, and agriculture. In the training process, GCA and the compared augmentation methods use the same number of training data, 10 088 in total, to ensure fairness.

2) *Implementation Details:* All DNNs used in the experiments are implemented with the PyTorch framework and executed on NVIDIA RTX 3090/4090 GPUs under

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.

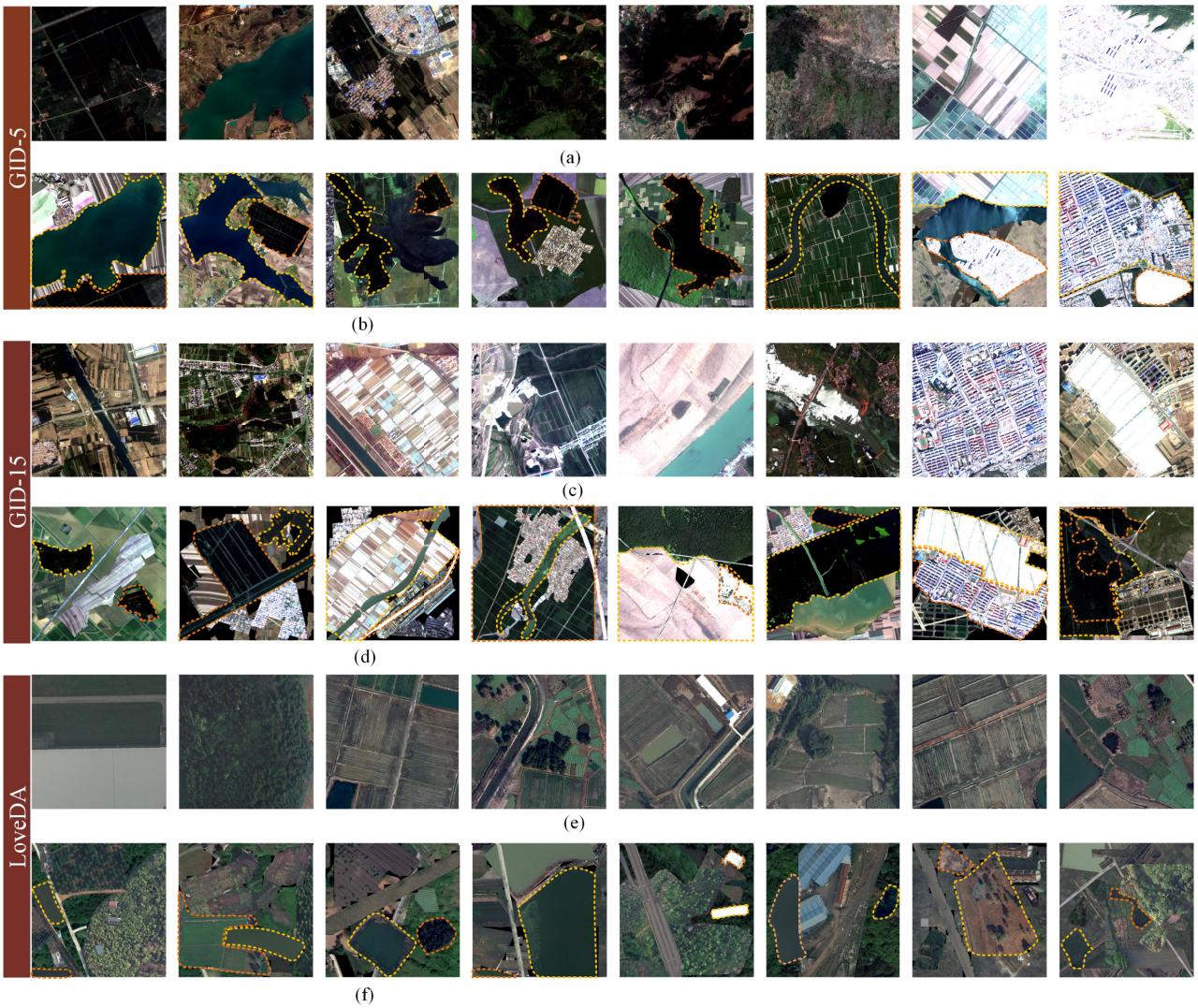


Fig. 8. Examples of the augmented training data generated by the GCA method and its corresponding raw data across the three datasets. (a), (c), and (e) Raw data. (b), (d), and (f) Augmented training data containing confused-class similar units.

CUDA 12.1. The optimizer is set to Adam, and a base learning rate is set to 0.01. A poly learning rate policy is employed in the training process, where the learning rate is multiplied by $(1 - (\text{iter}/\text{num_iter}))^{0.9}$ after each iteration [43]. The weight decay is set to 5×10^{-4}. All methods are trained with a batch size of 4. The number of training epochs is set to 50 for the GID dataset and 20 for the LoveDA dataset. To evaluate the performance of the proposed method, three metrics are used to measure experimental results, i.e., $F1$-score, overall accuracy (OA), and mean intersection over union (mIoU) [44].

B. Augmented Dataset

The proposed GCA method is applied to the three aforementioned semantic segmentation datasets, namely, GID-5, GID-15, and LoveDA. These datasets span diverse segmentation tasks, including coarse-grained land-cover segmentation, fine-grained semantic segmentation, and complex aerial scenes with multiscale objects. Details of the corresponding augmented datasets are as follows.

1) *GID-5 Data Augmentation:* To generate augmented training data containing confused-class similar units for coarse-grained semantic segmentation, the proposed GCA method is applied to the GID-5 dataset. Based on (6), four confused-class pairs (l, l') are identified: (*water*, *farmland*), (*built-up*, *farmland*), (*farmland*, *forest*), and (*forest*, *meadow*), as shown in Fig. 7(a).
2) *GID-15 Data Augmentation:* For fine-grained segmentation, 35 confused-class pairs (l, l') in GID-15 are identified based on (6), as listed in Fig. 7(c).
3) *LoveDA Data Augmentation:* For complex aerial scenes with multiscale objects, seven confused-class pairs (l, l') are identified based on (6): (*background*, *building*), (*background*, *road*), (*background*, *water*), (*background*, *barren*), (*background*, *forest*), (*water*, *agriculture*), and (*barren*, *agriculture*), as illustrated in Fig. 7(b).

These augmented datasets generated by GCA aim to highlight confused classes and guide the model to focus on discriminative features. To evaluate the performance of GCA, the number of confused-class pairs between the raw data and the augmented training data is compared on three datasets,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.

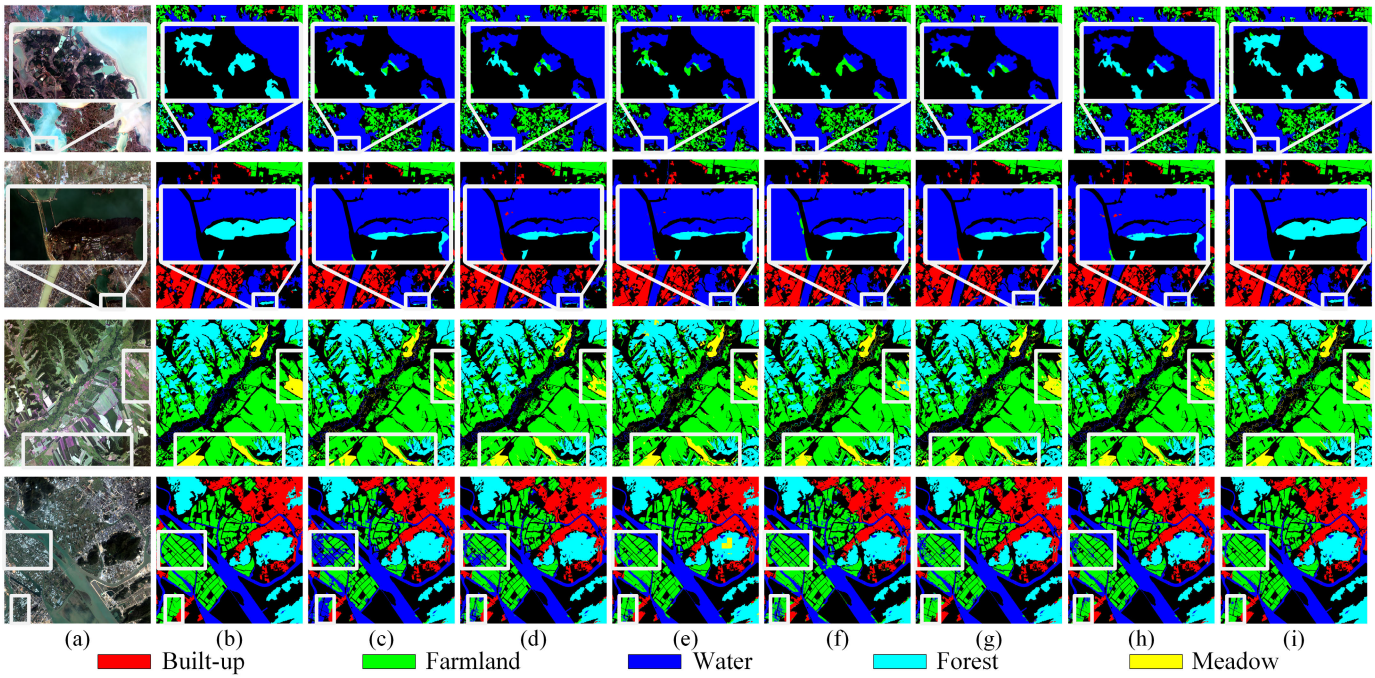


Fig. 9. Semantic segmentation results on GID-5 test images using models trained with different data augmentation methods. (a) Original remote sensing images; (b) ground truth; results from models trained on (c) raw data, (d) CutMix, (e) CBR, (f) cut-and-paste, (g) SRMF, (h) IPF-RAD, and (i) GCA.

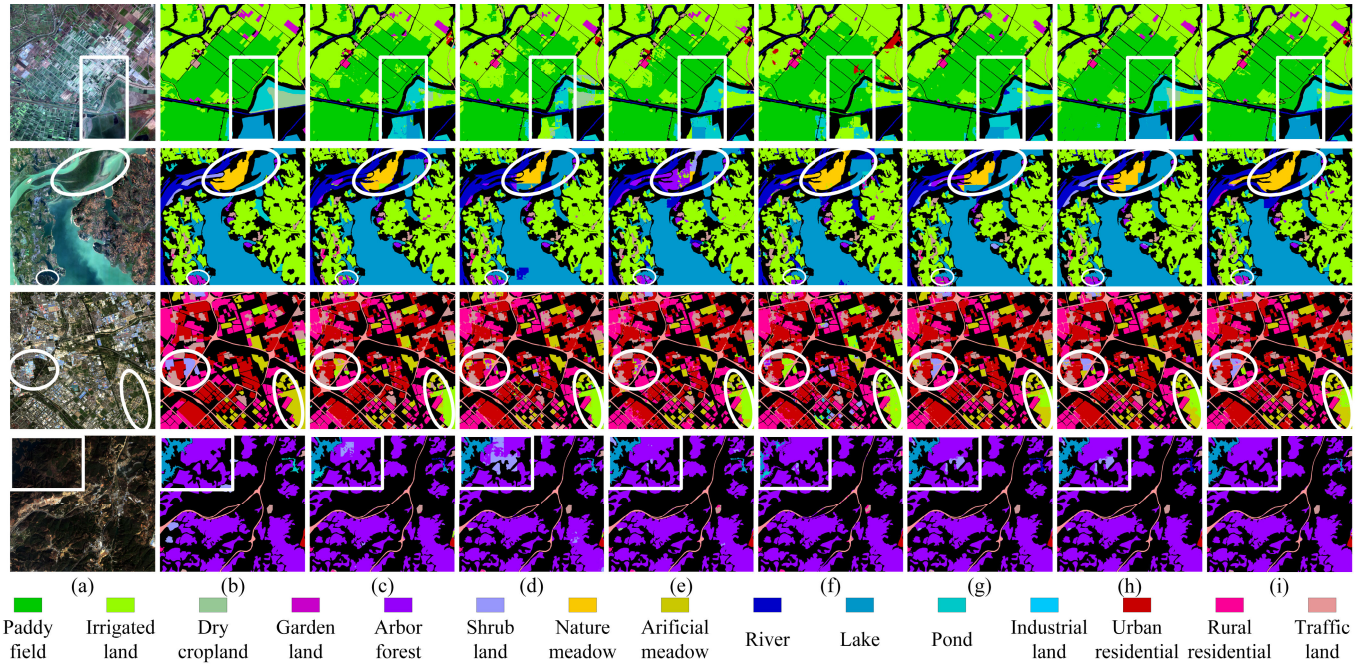


Fig. 10. Semantic segmentation results on the GID-15 test images using models trained with different data augmentation methods. (a) Original remote sensing images; (b) ground truth; results from models trained on (c) raw data, (d) CutMix, (e) CBR, (f) cut-and-paste, (g) SRMF, (h) IPF-RAD, and (i) GCA.

as shown in Fig. 7 and Table I. It can be observed that the augmented training data contain more confused-class pairs compared to the raw data. Moreover, these augmented training data introduce new confused-class pairs that are absent in the raw data, effectively enriching the challenging and hard-to-distinguish instances. In addition, the number of a few dominant confused-class pairs is lower in the augmented training data than in the raw data, such as (*background*, *building*) and (*paddy field*, *irrigated land*) in Fig. 7(b) and (c). This adjustment leads to a more balanced distribution of confused-class pairs and reduces the dominance of frequently occurring classes.

Visual examples of three datasets are shown in Fig. 8 to further illustrate the differences between the augmented training data and the raw data. As shown in Fig. 8, the augmented training data enhances confused-class pairs by

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.

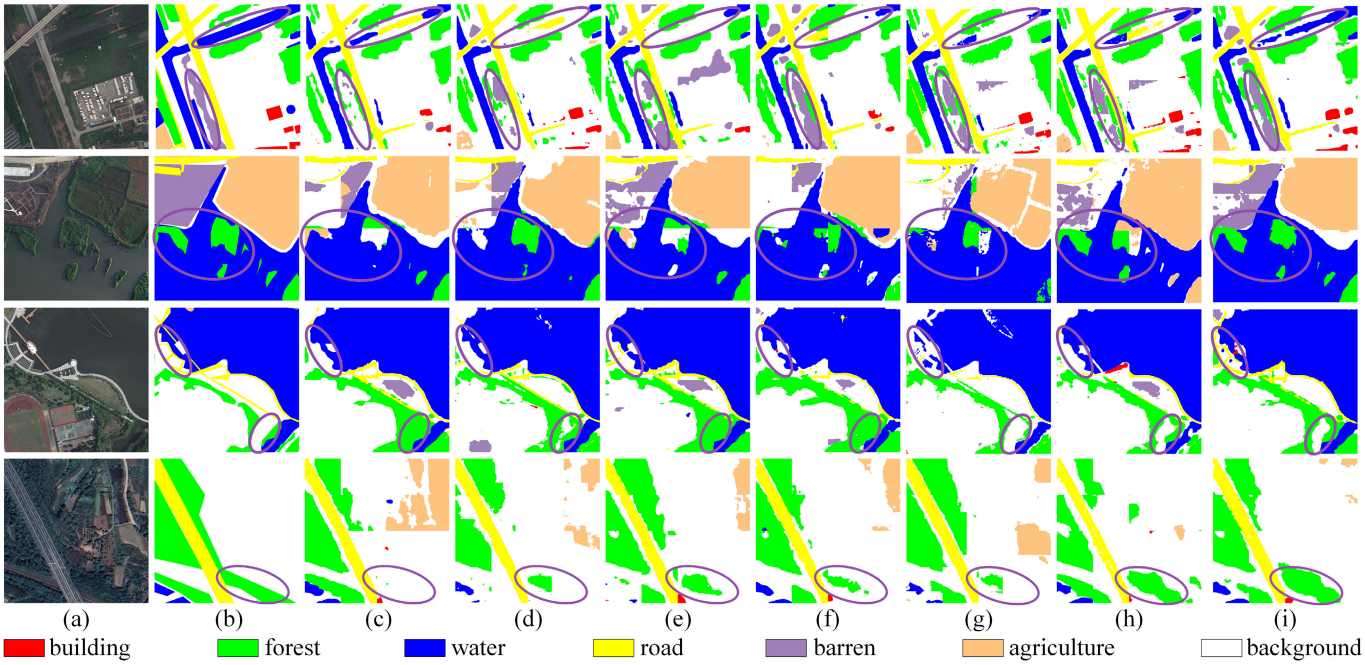


Fig. 11. Semantic segmentation results on the LoveDA test images using models trained with different data augmentation methods. (a) Original remote sensing images; (b) ground truth; results from models trained on (c) raw data, (d) CutMix, (e) CBR, (f) cut-and-paste, (g) SRMF, (h) IPF-RAD, and (i) GCA.

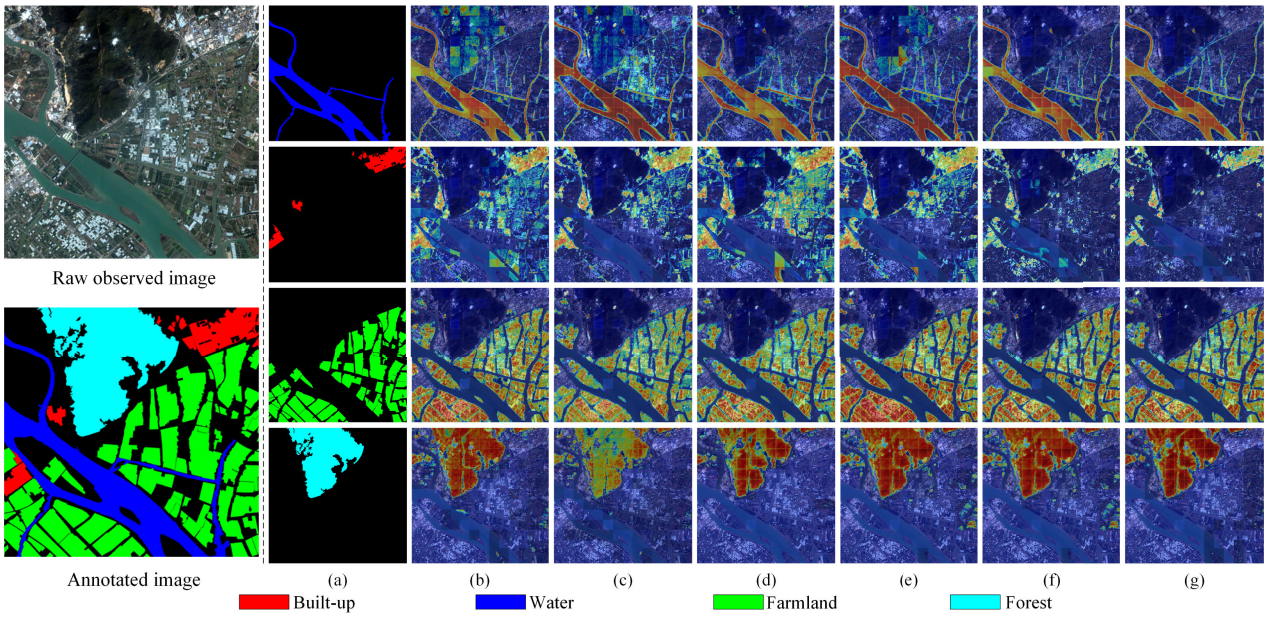


Fig. 12. Grad-CAM heatmaps under different configurations on the GID-5 dataset. (a) Annotated image of each class, (b) no GCA components, (c) GP, (d) (l, l'), (e) SR $+(l, l')$, (f) $(l, l') + (l, l')_C$, and (g) GCA method.

integrating spectrally similar units from different raw data. For example, the first training data in the second row combines such units from the first two raw data in the top row. By enhancing the diversity and occurrence of the confused-class pairs, the augmented training data will compel the DNN-based methods to focus on learning discriminative features between classes.

C. Experimental Results

In this section, we train different DNN-based methods on augmented training data generated by GCA for semantic segmentation. To evaluate the effectiveness of GCA, results with GCA are also compared with results trained on raw data and other state-of-the-art data augmentation methods on three datasets. Specifically, the employed data augmentation methods are as follows.

1) *CutMix [33]:* This method generates augmented data by replacing a random rectangular patch of an image with a patch from another image, which enriches both spatial and semantic diversity within the training data.
2) *CBR [38]:* This method enhances intraclass generalization by separating each class into independent maps

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.

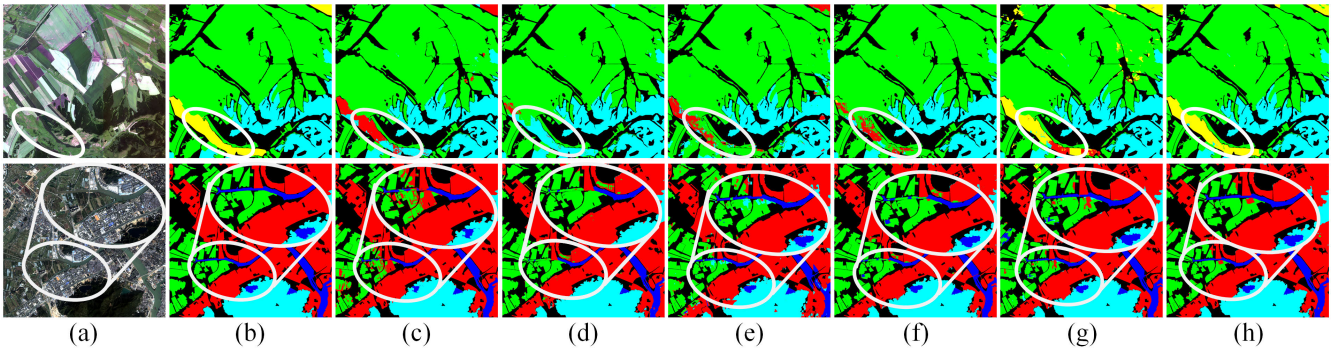


Fig. 13. Semantic segmentation results of each configuration on the GID-5 test images. (a) Original remote sensing image; (b) annotated image; results from models trained with (c) no GCA components, (d) GP, (e) (l, l'), (f) SR $+(l, l')$, (g) $(l, l') + (l, l')_C$, and (h) GCA method.

TABLE I

TOTAL NUMBER OF CONFUSED-CLASS PAIRS IN THE RAW DATA AND AUGMENTED TRAINING DATA ACROSS THE THREE DATASETS

Datasets	Total number	Total number of confused-class pairs	
		Raw data	Augmented training data
GID-5	3000	364	1175
GID-15	3000	419	1794
LoveDA	10088	7276	9774

and recombining single-class basic units to generate new training data, which strengthens the generalization information of intraclass.

3) *Cut-and-Paste [45]:* This method aims to enhance data diversity by extracting class-specific connected regions from segmentation masks and pasting them onto varied backgrounds, which enables region-level augmentation for satellite image segmentation.
4) *SRMF [46]:* This method performs multiscale cropping together with semantic reordering and resampling to generate tail-class-aware augmented training data, which alleviates the long-tail problem.
5) *IPF-RDA [47]:* This method introduces an information-preserving framework that identifies and preserves class-discriminative information during data augmentation, which enhances the robustness of augmented training data.

By comparing the performance of the above data augmentation methods, we aim to reveal their differences in exploiting the intrinsic information embedded within the data. To measure the robustness and generalization of GCA, various DNNs are tested, including BiSeNetV2 [48], SETR [49], Segmenter [50], kNet [51], Mask2Former [52], DDRNet [53], MAE [54], PoolFormer [55], EigNet [56], CgglNet [57], LSRFormer [58], and CIGFormer [59]. Among them, EigNet, CgglNet, LSRFormer, and CIGFormer are specifically designed for remote sensing image segmentation. To ensure the reliability and validity of the experimental results, all experiments are repeated five times, and the average performance is reported.

1) Experiments on the GID-5 Dataset: GID-5 is a large-scale land-cover classification dataset characterized by multitemporal properties and broad spatial distribution. These characteristics lead to high intraclass variability and low interclass separability, which pose typical challenges in remote sensing semantic segmentation. The quantitative results of GCA and the aforementioned data augmentation methods across different DNNs are listed in Table II. It can be observed that raw data exhibit considerable performance variability across different DNNs, with OA ranging from 89.42% to 97.41% and mIoU from 68.23% to 95.43%. Compared to the raw data, CutMix provides better performance by augmenting spatial contextual information, with average improvements of 1.2% in OA and 3.7% in mIoU. By leveraging the unique characteristics of remote sensing images, CBR achieves further improvements of 1.11% in OA and 2.58% in mIoU across different DNNs. Cut-and-paste further improves performance by augmenting instance-level diversity through object instance recomposition, achieving average gains of 1.50% in OA and 2.61% in mIoU across different DNN models. Beyond instance-level diversity, recent data augmentation methods have been proposed to explicitly address challenging samples and imbalanced class distributions to enhance remote sensing image segmentation performance. Specifically, SRMF achieves average improvements of 1.28% in OA and 4.30% in mIoU by alleviating long-tail class imbalance. IPF-RDA yields improvements of about 1.35% in OA and 4.61% in mIoU by preserving class-discriminative information during data augmentation. These results demonstrate that incorporating data-specific characteristics into data augmentation can improve model performance by better leveraging the inherent structure of the data. Building upon this, GCA alleviates interclass confusion by emphasizing highly similar units from confused-class pairs and achieves superior performance, with OA improvements ranging from 0.74% to 5.77% and mIoU gains from 1.28% to 16.65%. Furthermore, GCA exhibits improved stability, as reflected in lower standard deviations across various trials, demonstrating more robust and stable performance.

Example segmentation results of different data augmentation methods are shown in Fig. 9, and their quantitative indexes are given in Table III. From Fig. 9, the comparative methods exhibit many misclassification regions, particularly in areas with high spectral similarity, such as forest and water in the first two experiments and meadow and farmland in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.

TABLE II

QUANTITATIVE COMPARISON OF RAW DATA, CUTMIX, CBR, CUT-AND-PASTE, SRMF, IPF-RDA, AND GCA ON THE GID-5 DATASET ACROSS DIFFERENT DNN MODELS (UNIT: %). THE HIGHEST VALUES OF EVERY ROW ARE IN BOLD

Method	Raw data		CutMix		CBR		Cut-and-Paste		SRMF		IPF-RDA		GCA	
	OA	mIoU	OA	mIoU	OA	mIoU	OA	mIoU	OA	mIoU	OA	mIoU	OA	mIoU
BiSeNetV2	89.77 ±4.20	72.40 ±5.83	94.59 ±0.89	80.54 ±2.20	92.23 ±1.97	74.04 ±4.74	89.11 ±2.02	78.88 ±5.39	94.58 ±0.59	80.09 ±0.46	94.69 ±1.24	80.25 ±0.72	**95.54 ±0.59**	**81.12 ±0.30**
DDRNet	95.77 ±0.36	83.68 ±3.13	96.07 ±0.28	86.43 ±1.17	96.35 ±0.72	87.83 ±0.83	95.90 **±0.24**	83.24 ±2.01	96.43 ±0.54	89.56 ±1.40	96.34 ±0.67	89.49 ±0.71	**97.26 ±0.26**	**90.82 ±0.25**
MAE	96.35 ±0.16	87.56 ±0.64	96.64 **±0.06**	88.02 ±0.56	96.83 ±0.20	87.15 ±1.48	97.41 ±0.19	90.63 ±1.02	97.33 ±0.24	90.96 ±0.68	97.15 ±0.16	91.26 ±0.55	**98.31 ±0.18**	**92.61 ±0.51**
Mask2Former	94.49 ±0.38	73.53 ±5.14	95.56 ±0.73	82.97 ±5.84	96.03 ±0.33	85.36 ±0.82	95.62 ±0.94	71.12 ±2.88	96.96 ±0.43	88.04 ±0.75	97.07 ±0.43	88.28 ±0.97	**97.61 ±0.32**	**90.18 ±0.41**
Segmenter	96.06 ±0.38	87.69 ±0.49	95.88 ±0.32	87.82 ±0.29	96.18 ±0.29	82.69 ±1.34	96.92 ±0.45	86.36 ±2.32	96.61 ±0.74	86.67 ±0.95	96.76 ±0.38	87.17 ±0.52	**97.32 ±0.26**	**88.91 ±0.23**
SETR	96.33 ±1.08	87.42 ±2.34	97.47 ±0.79	89.86 ±2.52	97.75 ±0.68	91.28 ±1.35	97.68 ±0.59	90.83 ±1.50	97.32 ±0.68	91.15 ±0.83	97.61 ±0.41	91.95 ±0.89	**98.19 ±0.25**	**92.40 ±0.78**
CgglNet	97.39 ±0.15	90.90 ±0.55	97.53 ±0.11	90.31 ±1.92	97.85 ±0.12	89.59 ±1.02	97.44 ±0.61	**94.42** ±0.86	97.76 ±0.68	92.31 ±0.93	97.89 ±0.46	92.36 ±0.88	**98.41 ±0.10**	93.32 **±0.50**
LSRFormer	97.41 ±0.64	95.43 ±1.00	97.83 ±0.10	96.18 **±0.13**	97.68 ±0.58	96.06 ±0.88	98.02 ±0.09	95.90 ±1.36	97.89 ±0.16	95.56 ±0.78	98.09 ±0.15	96.11 ±0.38	**98.15 ±0.09**	**96.71** ±0.21
CIGFormer	97.40 ±0.39	90.58 ±1.18	97.71 ±0.72	95.38 ±0.55	97.19 ±0.17	95.26 ±0.83	97.43 ±0.29	89.89 ±4.57	97.59 ±0.33	93.56 ±0.74	97.48 ±0.49	93.82 ±0.83	**98.01 ±0.15**	**95.43 ±0.51**

TABLE III

QUANTITATIVE RESULTS ON THE GID-5 TEST IMAGES (UNIT: %). THE HIGHEST VALUES OF EVERY METRIC ARE IN BOLD

Images	Indexes	Raw data	CutMix	CBR	Cut-and-Paste	SRMF	IPF-RDA	GCA
GID5_1	Water	98.42	98.55	98.73	98.40	98.42	98.67	**99.82**
	Built-up	84.55	76.14	82.50	67.58	79.45	88.78	**97.50**
	Farmland	93.81	94.13	93.11	92.42	94.02	94.87	**97.52**
	Forest	56.03	58.45	39.28	46.46	57.61	56.68	**71.12**
	OA	98.00	97.80	98.08	97.59	97.97	98.11	**99.25**
	mIoU	83.20	78.40	81.82	76.21	82.37	84.01	**91.49**
GID5_2	Water	95.60	95.39	94.64	94.04	95.48	84.01	**99.34**
	Built-up	96.77	97.35	97.18	93.67	97.52	97.75	**99.02**
	Farmland	88.82	90.23	88.98	77.10	91.20	92.09	**96.06**
	Forest	25.39	26.34	26.69	25.65	25.74	38.18	**99.81**
	OA	97.33	97.27	97.55	94.56	97.60	97.77	**99.36**
	mIoU	61.32	76.87	77.32	72.61	80.34	82.12	**98.56**
GID5_3	Water	48.36	**69.10**	48.17	33.70	69.32	64.59	45.32
	Farmland	93.57	93.95	94.55	94.10	93.90	94.74	**97.63**
	Forest	95.22	95.63	93.17	91.07	94.92	95.52	**97.48**
	Meadow	62.78	65.47	62.08	62.71	64.29	65.73	**77.04**
	OA	95.19	95.85	95.14	94.56	95.53	96.08	**97.50**
	mIoU	59.98	**64.83**	59.59	50.25	64.49	64.11	60.61
GID5_4	Water	66.45	78.86	86.60	79.15	85.76	85.73	**94.04**
	Built-up	92.18	92.17	92.40	90.00	93.65	92.51	**97.70**
	Farmland	70.23	84.85	91.87	84.82	91.18	91.52	**95.76**
	Forest	95.21	95.31	92.29	94.89	95.43	95.51	**98.87**
	OA	88.23	93.15	94.90	92.87	95.30	95.39	**98.21**
	mIoU	64.81	72.63	87.80	87.21	91.18	91.32	**96.60**

the third. In contrast, the proposed GCA achieves the best visual performance, with results much closer to the ground truth. In addition, in the last experiment, water and farmland exhibit high spectral similarity and belong to the confused classes identified by GCA in this dataset. By augmenting such confused classes in the training data, GCA outperforms CBR, the best-performing method among the comparisons. Consequently, compared with the best-performing method, the IoU for the forest class improves from 58.45% to 71.12% in GID5_1 and from 38.18% to 99.81% in GID5_2. Likewise, the confused-class pair (water, farmland) exhibits substantial gains, with IoU rising from 86.60% to 94.04% and from 91.87% to 95.76% in GID5_4. These consistent improvements across confused-class pairs highlight the effectiveness of GCA in mitigating confusion between spectrally similar classes through targeted augmentation.

2) Experiments on the GID-15 Dataset: The GID-15 dataset refines the five coarse classes in the GID-5 into fifteen subclasses, significantly increasing class confusion and presenting a more challenging segmentation task. To evaluate the performance of GCA on fine-grained semantic segmentation, the GID-15 dataset is adopted. The quantitative evaluation results of the raw data and different data augmentation methods across various DNNs are shown in Table IV. Without any data augmentation, the raw data yield an average accuracy of 77.17%. Compared to the raw data, CutMix performs better. While CutMix improves the overall performance over the raw data, it exhibits inconsistent results across different DNNs, with degraded performance observed in certain models. This instability may be attributed to its lack of utilization of the unique characteristics of remote sensing data. In contrast, data augmentation approaches further enhance segmentation by leveraging specific characteristics of remote sensing images. Specifically, CBR enhances intraclass generalization by leveraging intraclass variability, while Cut-and-Paste increases sample diversity by inserting object instances into varied backgrounds. To address long-tail class distributions, SRMF incorporates a multiscale cropping technique together with a data augmentation strategy based on semantic reordering and resampling, and IPF-RDA emphasizes class-discriminative information by preserving critical semantics in the augmented training data. These methods consistently outperform the raw data. CBR and Cut-and-Paste achieve average OA gains of 1.70% and 1.50%, and average mIoU gains of 1.75% and 1.73%, respectively. Moreover, SRMF and IPF-RDA obtain OA improvements of 2.66% and 2.91%, and mIoU improvements of 4.36% and 5.64%, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

QUANTITATIVE COMPARISON OF RAW DATA, CUTMIX, CBR, CUT-AND-PASTE, SRMF, IPF-RDA, AND GCA ON THE GID-15 DATASET ACROSS DIFFERENT DNN METHODS (UNIT: %). THE HIGHEST VALUES OF EVERY ROW ARE IN BOLD

Method	Raw data		CutMix		CBR		Cut-and-Paste		SRMF		IPF-RDA		GCA	
	OA	mIoU	OA	mIoU	OA	mIoU	OA	mIoU	OA	mIoU	OA	mIoU	OA	mIoU
BiSeNetV2	65.60 ±2.06	27.20 ±2.32	68.19 ±3.16	29.03 ±1.35	69.41 **±0.70**	30.26 ±1.59	70.75 ±2.69	30.18 ±2.58	71.79 ±1.81	31.57 ±1.38	71.43 ±0.71	31.29 ±1.46	**72.12** ±1.51	**32.03** **±1.22**
DDRNet	74.46 ±2.93	37.75 ±2.04	76.24 ±1.43	38.61 ±0.72	77.44 ±1.39	40.62 ±1.43	76.62 ±0.75	42.51 ±1.82	77.06 ±0.72	41.36 ±0.95	77.50 ±0.87	41.86 ±0.53	**78.30** **±0.37**	**42.41** **±0.30**
MAE	77.39 ±5.92	42.47 ±8.66	68.10 ±0.64	32.33 ±0.61	81.00 ±0.28	46.39 ±0.49	82.64 ±0.24	47.31 **±0.20**	82.53 ±0.33	50.70 ±0.74	82.60 ±0.41	51.16 ±0.83	**83.10** **±0.20**	**52.64** ±0.69
Mask2Former	77.24 ±0.67	44.00 ±1.13	78.28 ±0.45	45.60 ±1.49	79.37 ±0.84	44.09 ±1.18	78.80 ±1.12	46.95 ±0.93	79.37 ±0.84	44.09 ±1.18	78.80 ±1.12	46.95 ±0.97	**80.63** **±0.38**	**48.57** **±0.91**
Segmenter	80.86 ±0.33	47.97 ±1.10	81.71 ±0.26	48.30 ±0.48	81.73 **±0.17**	47.87 ±0.68	82.14 ±1.03	47.84 ±1.16	81.82 ±0.54	47.90 ±1.08	82.12 ±0.28	50.23 ±1.08	**82.49** ±0.28	**51.60** **±0.42**
SETR	80.78 ±0.17	48.49 ±0.91	81.13 ±0.18	49.68 ±0.40	81.19 ±0.37	48.10 ±0.78	81.43 ±0.80	48.47 ±1.91	81.52 ±1.21	49.74 ±1.05	81.88 ±0.56	51.21 ±0.63	**82.86** **±0.11**	**53.29** **±0.39**
EigNet	80.63 ±0.72	43.66 ±2.88	81.08 ±0.12	48.89 ±1.25	82.84 ±0.37	50.12 ±0.93	79.80 ±1.88	45.92 ±2.34	83.36 ±1.03	57.72 ±1.23	84.12 ±0.36	57.32 ±0.94	**85.24** **±0.08**	**59.25** **±0.84**
CgglNet	82.00 ±1.37	52.55 ±0.99	82.08 ±0.94	51.30 ±1.12	82.51 ±0.44	52.46 ±0.48	83.56 ±0.89	53.26 ±0.91	83.45 ±0.79	56.80 ±1.45	84.93 ±0.68	58.12 ±1.01	**86.14** **±0.07**	**60.53** **±0.27**
LSRFormer	82.42 ±0.84	50.43 ±1.63	81.12 ±1.14	49.54 ±2.15	83.56 ±0.89	53.26 ±0.81	82.09 ±0.47	51.05 ±1.14	84.12 ±0.83	55.99 ±0.76	84.41 ±0.59	56.27 ±0.84	**85.32** **±0.41**	**58.00** **±0.57**
CIGFormer	80.56 ±1.34	50.47 ±1.09	82.36 ±1.26	53.88 ±1.98	79.86 ±2.04	49.34 ±3.16	80.14 ±2.13	49.76 ±3.28	83.55 ±0.86	56.74 ±1.24	84.23 ±0.64	56.98 ±1.21	**85.94** **±0.27**	**58.09** **±0.92**

TABLE V

QUANTITATIVE RESULTS ON THE GID-15 TEST IMAGES (UNIT: %). THE HIGHEST VALUES OF EVERY METRIC ARE IN BOLD

Images	Indexes	Raw data	CutMix	CBR	Cut-and-Paste	SRMF	IPF-RDA	GCA
GID15_1	paddy field	86.58	80.33	87.15	82.32	91.49	89.78	**93.88**
	irrigated land	81.50	74.45	77.13	71.95	85.26	83.68	**86.06**
	dry cropland	20.97	41.97	1.65	1.12	12.57	**23.16**	16.02
	garden land	57.28	58.10	41.97	2.27	57.22	51.03	**61.60**
	industrial land	9.58	8.57	6.19	1.79	21.25	**43.91**	10.40
	rural residential	44.62	37.27	43.06	29.03	40.34	29.45	**45.21**
	traffic land	38.43	40.83	38.50	29.69	**58.31**	57.32	51.05
	river	70.79	62.26	73.33	67.16	67.54	73.07	**74.12**
	OA	85.90	81.90	84.45	79.53	88.00	89.50	**91.74**
	mIoU	43.34	40.33	33.69	27.92	38.57	39.00	**53.61**
GID15_2	irrigated land	95.64	96.57	96.41	93.67	96.20	97.96	**96.69**
	Garden land	19.22	31.72	28.30	-	**74.05**	40.98	37.73
	Arbor forest	48.68	51.06	17.42	34.83	36.27	52.29	**71.38**
	Shrub land	-	-	0.02	0.07	26.25	**64.10**	2.89
	Nature meadow	84.49	65.00	7.29	**85.97**	59.10	71.47	81.72
	Rural residential	74.20	81.38	85.85	84.19	88.56	81.90	**89.35**
	River	69.61	68.97	68.79	57.56	**83.24**	69.14	75.35
	Lake	88.61	85.99	87.03	87.78	89.04	88.37	**89.48**
	Pond	37.01	34.27	37.06	21.04	39.39	34.12	**43.54**
	OA	91.54	90.96	87.65	89.31	**93.56**	92.77	93.02
	mIoU	40.66	41.16	30.59	33.22	41.03	40.69	**49.01**
GID15_3	garden land	44.21	22.89	27.18	-	50.21	50.52	**50.94**
	shrub land	0.00	5.26	13.27	0.11	55.57	57.03	**64.33**
	artificial meadow	55.51	47.94	67.80	9.12	66.41	69.38	**81.24**
	industrial land	72.21	70.43	63.89	65.11	69.20	70.14	**72.84**
	urban residential	**64.01**	61.99	58.04	56.50	62.10	63.01	62.53
	rural residential	63.03	68.01	63.68	41.38	72.12	69.18	**69.22**
	traffic land	62.77	59.12	63.33	63.69	65.66	**71.75**	66.13
	OA	76.50	74.43	74.38	64.14	78.30	79.01	**81.22**
	mIoU	30.14	27.98	27.52	21.47	30.14	30.11	**38.85**
GID15_4	arbor forest	94.73	89.23	95.58	90.82	95.38	94.67	**95.92**
	industrial land	68.21	72.23	88.69	**90.73**	85.66	86.34	79.74
	urban residential	55.04	54.11	36.14	22.47	68.92	78.13	**73.51**
	rural residential	84.06	84.78	66.18	44.96	86.00	89.81	**89.85**
	traffic land	94.92	86.53	**95.34**	76.57	84.84	94.28	90.65
	OA	93.87	89.63	95.16	90.54	95.62	94.78	**95.94**
	mIoU	43.84	43.87	40.07	35.28	37.70	36.66	**52.07**

However, they mainly focus on intraclass variability and data distribution characteristics, while they do not account for confusion between different classes. By increasing the occurrence of confused-class pairs, GCA consistently outperforms other data augmentation methods across different DNNs, achieving a maximum improvement of 6.52% in OA and 15.59% in mIoU, with average gains of 4.02% and 6.14%, respectively. Moreover, GCA also demonstrates improved stability, with an average variation of 0.37% in OA and 0.65% in mIoU across all DNNs, significantly lower than those of other methods.

Fig. 10 provides a qualitative comparison of the segmentation results produced by different data augmentation methods, and the corresponding quantitative results are presented in Table V. It can be observed that class subdivision leads to a significant increase in confusion between different classes. Typical examples include a lake and pond in the first experiment, garden land and arbor forest in the second, artificial meadow and irrigated land in the third, and arbor forest and shrub land in the final experiment. This spectral similarity leads to extensive misclassification in the comparative methods. In contrast, the proposed GCA method achieves the best visual performance by balancing the distribution of confused classes and explicitly emphasizing them during training. The quantitative results in Table V further confirm the advantage of GCA. Specifically, GCA achieves the highest mIoU across all test images. Compared with the best competing methods, GCA yields OA improvements of up to 2.24% (on GID15_1) and significant mIoU gains ranging from 7.85% to 10.27%.

3) Experiments on the LoveDA Dataset: To further test the generalization performance of the proposed GCA method, we compare it with other data augmentation methods on the LoveDA dataset. The LoveDA dataset is more challenging than GID, as its higher spatial resolution introduces greater class confusion and requires more precise feature discrimination. The numerical results of the raw data, other data augmentation methods, and GCA across different DNNs are shown in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

QUANTITATIVE COMPARISON OF RAW DATA, CUTMIX, CBR, CUT-AND-PASTE, SRMF, IPF-RDA, AND GCA ON THE LOVEDA DATASET ACROSS DIFFERENT DNN METHODS (UNIT: %). THE HIGHEST VALUES OF EVERY ROW ARE IN BOLD

Method	Raw data		CutMix		CBR		Cut-and-Paste		SRMF		IPF-RDA		GCA	
	OA	mIoU	OA	mIoU	OA	mIoU	OA	mIoU	OA	mIoU	OA	mIoU	OA	mIoU
DDRNet	63.81 ±0.77	44.99 ±0.67	64.81 ±0.58	46.87 ±0.62	65.05 ±0.78	45.98 ±0.94	64.30 ±1.31	44.13 ±1.74	65.18 ±1.13	46.21 ±0.77	65.31 ±0.58	46.00 ±0.69	**66.18 ±0.57**	**47.85 ±0.55**
KNET	66.34 ±1.61	48.08 ±0.82	66.65 ±0.78	48.12 ±0.62	67.89 **±0.25**	48.89 ±0.78	66.69 ±1.45	46.91 ±1.87	67.72 ±0.90	48.89 ±0.99	67.96 ±1.05	49.50 ±0.96	**68.66** ±0.77	**50.50 ±0.60**
Mask2Former	67.45 ±1.58	49.31 ±1.65	60.39 ±0.48	36.57 ±1.00	68.47 ±0.22	48.70 **±0.54**	67.89 ±0.90	48.66 ±0.77	68.49 ±0.27	48.86 ±0.84	68.76 ±0.36	49.06 ±0.99	**69.44 ±0.22**	**50.30** ±0.89
PoolFormer	66.05 ±0.67	47.69 ±0.62	66.24 ±0.49	47.55 ±0.50	66.63 ±0.54	47.53 ±0.47	61.27 ±3.64	44.90 ±1.43	66.60 ±0.48	47.19 ±0.86	66.81 ±0.94	47.78 ±0.87	**67.85 ±0.46**	**48.88 ±0.35**
SegFormer	68.01 ±0.21	49.53 ±0.25	66.84 ±0.31	46.20 ±0.38	68.40 ±0.24	49.76 ±0.27	**69.76** ±1.26	51.26 ±0.25	68.65 ±0.98	50.27 ±0.26	68.95 ±0.56	50.86 ±0.42	69.37 **±0.09**	**51.39 ±0.24**
SETR	70.52 ±0.34	52.00 ±0.26	70.86 ±0.21	**53.00** ±0.36	70.36 ±0.23	51.45 ±0.33	70.96 ±0.31	51.59 ±0.35	70.51 ±0.49	52.10 ±0.63	70.89 ±0.41	51.89 ±0.61	**71.23 ±0.20**	52.59 **±0.30**
EigNet	65.72 ±2.83	48.26 ±1.52	66.59 ±0.80	48.01 ±0.70	66.91 ±1.04	47.90 ±0.74	67.70 ±1.10	45.56 ±2.05	67.82 ±1.22	48.34 ±1.13	68.12 ±0.97	48.95 ±0.58	**68.92 ±0.55**	**49.49 ±0.21**
CgglNet	65.92 ±2.97	48.39 ±1.61	67.18 ±0.31	47.67 ±0.39	67.76 ±0.59	48.35 ±0.40	68.66 ±1.22	47.80 ±1.90	68.60 ±0.71	49.20 ±0.74	68.90 ±0.65	49.78 ±0.63	**69.76 ±0.31**	**50.77 ±0.38**
LSRFormer	70.79 ±0.29	51.84 ±0.74	71.02 ±0.95	49.98 ±1.03	71.31 ±0.57	52.19 ±0.52	71.30 ±2.68	49.98 ±6.75	71.43 ±0.69	50.84 ±0.42	71.68 ±0.57	52.06 ±0.38	**72.28 ±0.14**	**53.03 ±0.36**
CIGFormer	67.27 ±1.58	46.67 ±1.40	68.04 ±0.60	48.85 **±0.58**	68.31 ±2.20	49.18 ±1.41	63.43 ±3.26	47.00 ±2.19	67.94 ±0.51	48.43 ±0.95	68.39 ±0.68	48.97 ±0.89	**68.84 ±0.50**	**49.46** ±0.70

TABLE VII

QUANTITATIVE RESULTS ON THE LOVEDA TEST IMAGES (UNIT: %). THE HIGHEST VALUES OF EVERY METRIC ARE IN BOLD

Images	Indexes	Raw data	CutMix	CBR	Cut-and-Paste	SRMF	IPF-RDA	GCA
LoveDA_1	Background	69.82	69.97	63.02	67.04	70.10	71.56	**71.57**
	Building	35.48	**59.94**	16.16	25.07	57.81	55.30	54.79
	Road	65.76	68.41	75.86	68.25	**79.14**	73.45	74.14
	Water	50.06	42.96	45.72	44.12	49.63	52.09	**59.51**
	Barren	5.63	25.49	23.21	51.90	43.99	40.83	**55.06**
	Forest	**59.63**	56.89	55.73	58.38	56.16	57.57	56.67
	Agriculture	2.14	54.90	60.80	4.33	81.46	63.89	**81.78**
	OA	76.52	77.01	72.93	75.57	78.58	79.03	**80.13**
	mIoU	41.22	54.08	48.64	45.58	62.91	59.24	**64.79**
LoveDA_2	Background	6.78	5.75	5.67	6.27	7.94	9.31	**16.80**
	Road	50.94	43.22	**51.32**	34.93	48.19	36.85	36.83
	Water	89.95	91.10	91.81	88.42	87.62	91.74	**91.90**
	Barren	29.83	27.82	56.43	17.73	25.30	44.56	**63.11**
	Forest	34.43	57.72	41.58	43.02	32.88	42.60	**78.39**
	Agriculture	80.56	78.42	78.33	**87.56**	77.53	85.48	82.56
	OA	76.64	77.38	80.15	76.31	72.39	80.18	**84.06**
	mIoU	48.75	43.43	54.19	39.70	43.58	50.09	**59.83**
LoveDA_3	Background	74.02	78.45	74.79	70.99	79.32	79.89	**80.21**
	Road	44.97	44.37	43.99	28.24	23.47	28.66	**51.47**
	Water	96.05	96.25	96.61	95.91	92.46	96.12	**96.87**
	Forest	48.03	54.23	47.32	41.84	**62.97**	58.98	50.90
	OA	87.29	89.41	87.59	85.34	89.52	89.65	**89.94**
	mIoU	43.84	39.04	37.53	33.85	36.89	37.67	**39.85**
LoveDA_4	Background	65.68	74.02	75.37	78.82	72.72	79.01	**82.98**
	Road	77.71	75.63	75.93	76.12	79.76	66.49	**82.41**
	Water	73.84	46.84	9.20	74.17	48.40	71.26	**79.64**
	Forest	60.99	51.13	61.59	64.36	62.19	57.51	**65.89**
	OA	74.26	79.63	81.42	83.87	79.95	83.90	**87.51**
	mIoU	46.37	41.27	37.01	48.91	50.56	45.71	**51.82**

TABLE VIII

QUANTITATIVE RESULTS UNDER DIFFERENT θ VALUES (UNIT: %). THE HIGHEST VALUES FOR EACH METRIC ARE BOLDED

Parameter θ	Water	Built-up	Farmland	Forest	Meadow	OA	mIoU
0.9	94.89	95.20	93.93	95.47	24.01	97.03	80.70
0.8	94.52	97.18	95.22	93.16	62.36	97.49	**89.78**
0.7	95.58	97.30	95.58	92.64	52.02	**97.63**	88.81
0.6	94.39	96.69	94.73	93.88	57.13	97.31	87.57
0.5	95.26	94.63	94.26	93.93	50.75	97.10	85.77
0.4	93.84	89.22	90.58	78.82	1.74	94.48	70.84

TABLE IX

ADDITIONAL TIME COST (S) OF GENERATING AUGMENTED TRAINING DATA FOR DIFFERENT DATA AUGMENTATION METHODS

	CutMix	CBR	Cut-and-Paste	GCA
GID-5	0.1	6	5	5
GID-15	0.1	10	5	11
LoveDA	0.1	15	5	15

TABLE X

ABLATION RESULTS OF DIFFERENT COMPONENT COMBINATIONS ON THE GID-5 DATASET (UNIT: %). THE HIGHEST VALUES FOR EACH METRIC ARE BOLDED

(l, l')	$(l, l')_C$	GP	OA	mIoU
–	–	–	93.08	71.86
–	–	✓	93.90	73.68
✓	–	–	94.08	74.15
✓	–	✓	94.50	74.77
✓	✓	–	96.26	87.04
✓	✓	✓	**98.20**	**93.99**

Table VI. Compared to the raw data, CutMix, CBR, Cut-and-Paste, SRMF and IPF-RDA achieve maximum improvements of 1.26%, 1.84%, 2.74%, 2.70%, and 3.48% in OA, and 2.18%, 2.51%, 1.73%, 2.56%, and 2.51% in mIoU, respectively. However, these methods overlook the impact of class confusion caused by high spectral similarity. By augmenting confused classes, GCA outperforms these methods, achieving an average improvement of approximately 0.47% in OA and 0.59% in mIoU over the best-performing method.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.

TABLE XI
CONFUSED-CLASS PAIRS OBTAINED FROM DIFFERENT PRESEGMENTATION MODELS

Confused-class	Mask2Former	MAE	SegFormer
(water, farmland)	✓	✓	✓
(built-up, farmland)	✓	✓	✓
(farmland, forest)	-	✓	-
(forest, meadow)	-	-	✓

TABLE XII
QUANTITATIVE COMPARISON ACROSS DIFFERENT INITIAL MODELS (UNIT: %)

Model	Water	Built-up	Farmland	Forest	Meadow	OA	mIoU
Mask2Former	92.02	97.08	96.06	98.12	77.99	97.80	92.25
MAE	94.22	97.01	96.71	98.56	79.05	98.17	93.11
Segmenter	93.83	97.18	96.69	98.19	80.09	98.13	92.78

Segmentation examples are presented in Fig. 11, and their corresponding numerical results are reported in Table VII. In these experiments, class confusion becomes more prominent. In the first and third experiments, the barren regions highlighted by gray ellipses are often misclassified as background due to their spectral similarity. In the second experiment, water and forest exhibit similar spectral characteristics, leading to confusion between the two classes. Similarly, in the last experiment, forest areas show high similarity to surrounding background regions, making them difficult to distinguish using existing data augmentation methods. In contrast, GCA effectively reduces such confusion, resulting in more precise segmentation results. The results in Table VII further indicate that GCA achieves the largest performance gains for confused-class pairs, while also maintaining leading OA. These results validate the effectiveness of GCA in improving class separability and addressing highly confusable classes.

D. Parameter Analysis of θ

The proposed GCA method aims to enhance the learning of discriminative features within easily confused classes by increasing their occurrence in the training data. To achieve this, parameter θ is designed to identify confused-class similar units, ensuring that only obvious confused-class pairs are incorporated into the augmentation process. But if θ is set too high, only a few highly similar-unit pairs will be selected, limiting the diversity of augmented training data. Conversely, if θ is set too low, geographic units with low similarity may be included, reducing the effectiveness of the confused-class emphasis. Therefore, an appropriate θ is essential for maintaining a proper balance between the diversity and similarity of geographic units in GCA.

To determine the value of θ, we tested θ in the range of [0.4, 0.9] with a step of 0.1 on the GID-5 dataset, and the corresponding quantitative results are shown in Table VIII. It can be seen that, when θ exceeds 0.8, the performance of the proposed method decreases noticeably, which can be attributed to the limited diversity caused by repeatedly selecting a small number of highly similar units. Conversely, when θ drops below 0.5, the quantitative metrics also decline, since including insufficiently similar units weakens the method's ability to capture representative features within each land category. Therefore, the appropriate interval is $[0.6, 0.8]$, and the value of θ sets to 0.8 for all experiments.

E. Additional Time Cost Comparison

Time cost is also an important metric for evaluating data augmentation methods. We report the additional computational cost introduced at the data augmentation stage by GCA and other methods. In GCA, Modules *G* and *C* involve a one-time computation for geospatial prior extraction and confused-class similar-unit identification, requiring 1.25, 1.85, and 1.17 h on the GID-5, GID-15, and LoveDA datasets, respectively. As these results are reusable, we further compare the time required by Module *A* to generate augmented training data with that of other augmentation methods in Table IX. Module *A* introduces a manageable computational overhead. Unlike simpler methods such as CutMix, GCA generates augmented samples designed to alleviate interclass confusion and promote intraclass diversity. Moreover, the augmented training data are generated during the preprocessing stage, GCA imposes no additional computational burden during model training or optimization.

F. Ablation Study

To demonstrate the effectiveness of each key component in the proposed GCA method, we conduct an ablation study on the GID-5 dataset. This experiment compares six different configurations, where the components include: (l, l') indicates confused-class pair probability enhancement, $(l, l')_C$ refers to similar units augmentation, and GP represents geospatial prior guided training data generation. The configurations are as follows.

1) *–:* All GCA components are disabled.
2) *GP:* Training data are generated by integrating randomly sampled geographic units under the guidance of geospatial prior information.
3) (l, l')*:* Training data are generated by sampling geographic units according to the enhanced probability of confused-class pairs.
4) *GP* $+(l, l')$*:* Training data are generated by sampling geographic units according to the enhanced probability of confused-class pairs and integrating them under the guidance of geospatial prior information.
5) $(l, l') + (l, l')_C$*:* Training data are generated by augmenting confused-class similar units without considering geospatial prior information.
6) *GP* $+(l, l') + (l, l')_C$*:* The proposed GCA method, incorporating all components.

The quantitative performance of each configuration is presented in Table X. Compared with the configuration without GCA, adding the GP component improves OA and mIoU by 0.82% and 1.82%, respectively, while adding the (l, l') component yields gains of 1.0% in OA and 2.19% in mIoU. These results indicate that both geospatial priors and confused-class information provide effective guidance for generating training

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.

TABLE XIII

QUANTITATIVE COMPARISON BETWEEN THE PROPOSED GCA METHOD AND OTHER DATA AUGMENTATION METHODS ON THE GID-5 DATASET UNDER DIFFERENT GPU DEVICES (UNIT: %). THE HIGHEST VALUES OF EVERY ROW ARE IN BOLD

GPU	Method	Raw data		CutMix		CBR		GCA	
		OA	mIoU	OA	mIoU	OA	mIoU	OA	mIoU
RTX3080Ti	Mask2Former	95.81	74.32	96.86	86.84	96.83	87.97	**97.38**	**89.67**
RTX3090	Mask2Former	95.78	78.89	95.97	96.34	96.05	84.91	**98.14**	**93.81**
RTX5000	Mask2Former	95.07	69.38	89.91	83.92	96.40	88.85	**97.11**	**88.59**
RTX6000	Mask2Former	95.48	72.08	96.53	79.91	96.72	88.85	**97.64**	**90.32**
RTX8000	Mask2Former	95.39	75.01	95.67	73.68	96.33	81.24	**97.60**	**89.42**

TABLE XIV

QUANTITATIVE COMPARISON OF THE PROPOSED GCA METHOD AND OTHER AUGMENTATION METHODS UNDER DIFFERENT RANDOM DATA SPLITS ON THE GID-5 DATASET (UNIT: %). THE TRAINING/TESTING SPLIT RATIO IS FIXED, BUT THE SAMPLES ARE RANDOMLY CHANGED ACROSS RUNS. THE HIGHEST VALUE IN EACH ROW IS HIGHLIGHTED IN BOLD

Indexes	Method	Raw data		CutMix		CBR		GCA	
		OA	mIoU	OA	mIoU	OA	mIoU	OA	mIoU
1	Mask2Former	95.36	73.93	95.89	76.78	96.51	80.44	**97.26**	**85.45**
2	Mask2Former	93.04	81.76	91.82	79.61	95.41	88.37	**96.14**	**90.97**
3	Mask2Former	86.84	71.88	90.21	79.96	90.16	83.03	**96.70**	**91.14**

data. Building on these individual components, combining GP with (l, l') yields further improvements, achieving an additional 0.60% in OA and 1.09% in mIoU over the GP configuration. Meanwhile, the $(l, l') + (l, l')_C$ configuration yields a greater improvement over the configuration without any GCA components, achieving improvements of 3.18% in OA and 15.18% in mIoU. These results validate that both confused-class pair probability enhancement and confused-class similar-unit augmentation can effectively improve performance by augmenting easily misclassified units between highly confusable classes. By integrating all components, GCA achieves the highest gains in both OA and mIoU, further confirming its effectiveness in learning discriminative features. In addition, gradient-weighted class activation map (Grad-CAM) [60] is used to visualize and evaluate the discriminative capability of different configurations, as shown in Fig. 12. In this experiment, farmland and water (or built-up) exhibit highly similar spectral characteristics, leading most configurations to activate large overlapping regions and produce ambiguous boundaries. In contrast, the configuration incorporating both (l, l') and $(l, l')_C$ can generate more compact and precisely aligned activation regions for built-up than configurations without these components, as shown in Fig. 12(f) and (g). Meanwhile, the GP component helps mitigate incorrect spatial responses, such as misleading activations of built-up within water regions. The activation maps of GCA are noticeably cleaner, with attention concentrated on class-relevant regions rather than irrelevant regions. This indicates that GCA can promote discriminative feature learning more effectively, enabling the model to focus on subtle interclass differences instead of simply amplifying activation responses. The qualitative comparison in Fig. 13 further illustrates the visual improvements achieved by different component configurations. These results demonstrate the effectiveness of each component of GCA.

IV. DISCUSSION

In this section, we further discuss the proposed GCA method from three additional perspectives. First, we evaluate the impact of confusion matrices computed from different presegmentation models. Second, we test its stability by comparing the performance of GCA with other data augmentation methods across different GPU devices. Third, to evaluate the robustness of GCA under varied training and testing image compositions, we exchange the images between the training and testing sets, while keeping the number of images in each set unchanged.

A. Stability Analysis Under Different Presegmentation Models

In the proposed GCA, the identification of confused-class pairs is essential because it determines which classes are emphasized during augmentation and directly guides the generation of augmented samples. These pairs are derived from the confusion matrix computed by DNN-based presegmentation results, and different initial models may yield different confused classes. To evaluate the impact of the initial model on GCA, three different DNN-based networks are selected to compute their corresponding confusion matrices and generate confused-class pairs, the results are shown in Table XI. As shown in Table XII, the resulting performance varies only slightly, with mIoU values ranging from 92.25% to 93.11%, indicating that GCA is relatively stable with respect to the choice of the initial model.

B. Stability Evaluation Under Different GPUs

To evaluate the performance of the GCA with various GPU devices, we conduct experiments on RTX 3080 Ti, 3090, 5000, 6000, and 8000, respectively. Table XIII lists the quantitative results of GCA and other data augmentation methods on different GPUs using Mask2Former. It can be observed that GCA consistently outperforms other data augmentation methods across all tested GPUs. Specifically, GCA achieves an average improvement in OA of 2.07%, 1.85%, and 1.11%, and in mIoU of 15.75%, 4.46%, and 3.38%, compared to raw data, CutMix, and CBR, respectively. This consistent performance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.

across diverse GPUs demonstrates the stability of GCA and further validates its effectiveness.

C. Robustness Evaluation Under Different Random Data Splits

To test the robustness of the proposed GCA method, we conduct comparative experiments on the GID-5 dataset under various random training/testing splits with a fixed ratio. Specifically, three independent rounds of data splitting are performed, in which the training and testing sets are randomly reassigned while maintaining the same split ratio. The numerical results are presented in Table XIV. As shown, the performance of other augmentation methods varies significantly across different splits, whereas GCA consistently yields more stable and superior results. These results demonstrate the robustness of GCA.

V. Conclusion

In this article, we propose a novel confused-class data augmentation (GCA) method for the semantic segmentation of remote sensing images. GCA aims to improve class separability by augmenting confused classes under the guidance of geospatial prior information. The strength of GCA lies in two main aspects. First, GCA is designed based on the unique characteristics of remote sensing images, aiming to compel the model to learn discriminative features and to reduce misclassification caused by spectral variability and high interclass similarity. Second, augmented training data are generated by integrating highly similar units from confused classes under the guidance of geospatial prior information.

The GCA method comprises three key modules. Module G extracts geographic units and spatial relationships as geospatial prior information to guide the generation of augmented training data. Module C identifies the collection of similar units from a confused-class pair. And module A generates the training data by augmenting confused-class similar units under the guidance of geospatial relationships. Experimental results demonstrate the effectiveness, generalization, and robustness of GCA across different DNNs and datasets, compared to other state-of-the-art data augmentation methods. The future work will be focused on further improving GCA from three aspects: 1) diffusion models can be incorporated to synthesize diverse and realistic confused-class similar units for use in augmented training data; 2) augmentation strategies can be considered specifically for underrepresented confused classes to address class imbalance in the training data; and 3) the proposed GCA method is used for multiclass segmentation, but when applied to binary segmentation, the identification of interclass confusion becomes less informative due to the presence of only two classes, hence, our future work will also further consider instance-level augmentation for binary segmentation.

Acknowledgment

The authors are very grateful to the editors and the anonymous referees for their constructive and valuable comments and suggestions, which led to the present greatly improved version of their manuscript.

References

[1] X. Ye, P. Wang, J. Zhu, Y. Duan, and B. Yang, "Land Surface Temperature end-to-end retrieval considering the topographic effect using radiative transfer model-driven convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5001010.

[2] Y. Xing et al., "MAM-YOLOv9: A multiattention mechanism network for methane emission facility detection in high-resolution satellite remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5614516.

[3] T. Mu, Q. Zheng, and S. Y. He, "Robust disaster impact assessment with synthetic control modeling framework and daily nighttime light time series images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4400712.

[4] N. Zhou et al., "Submeter-level global-scale road extraction based on limited labeled data from optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4411323.

[5] Y. Cai, Y. Yang, Y. Shang, Z. Shen, and J. Yin, "DASRSNet: Multitask domain adaptation for super-resolution-aided semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2022, Art. no. 5600218.

[6] J.-J. Huang et al., "DeMPAA: Deployable multi-mini-patch adversarial attack for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5623613.

[7] H. Chen, W. Yang, L. Liu, and G.-S. Xia, "Coarse-to-fine semantic segmentation of satellite images," *ISPRS J. Photogramm. Remote Sens.*, vol. 217, pp. 1–17, Nov. 2024.

[8] M. Otsu, J. Zhou, B. Han, A. Accad, and S. Pan, "A robust co-training framework to handle noisy labels for remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 3000814.

[9] W. Xu, J. Wang, Z. Wei, M. Peng, and Y. Wu, "Deep semantic-visual alignment for zero-shot remote sensing image scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 198, pp. 140–152, Apr. 2023.

[10] W. Wang, S. Brönnimann, J. Zhou, S. Li, and Z. Wang, "Near-surface air temperature estimation for areas with sparse observations based on transfer learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 220, pp. 712–727, Feb. 2025.

[11] R. Thoreau, L. Risser, V. Achard, B. Berthelot, and X. Briottet, "Toulouse hyperspectral data set: A benchmark data set to assess semi-supervised spectral representation learning and pixel-wise classification techniques," *ISPRS J. Photogramm. Remote Sens.*, vol. 212, pp. 323–337, Jun. 2024.

[12] Z. Zhang, T. Zhao, Y. Guo, and J. Yin, "RS5M and GeoRSCLIP: A large-scale vision{-} language dataset and a large vision-language model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5642123.

[13] H. Zhang, N. Liu, H. Wu, and J. Huo, "Stratigraphic sequence correlation of well logs using KANFormer enhanced with OMP-based data augmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4503011.

[14] S. Yang, P. Li, X. Xiong, F. Shen, and J. Zhao, "AdaAugment: A tuning-free and adaptive approach to enhance data augmentation," *IEEE Trans. Image Process.*, vol. 34, pp. 4843–4855, 2025.

[15] H. Yar, W. Ullah, Z. A. Khan, and S. W. Baik, "An effective attention-based CNN model for fire detection in adverse weather conditions," *ISPRS J. Photogramm. Remote Sens.*, vol. 206, pp. 335–346, Dec. 2023.

[16] R. Wang, Y. Yao, P. Chen, L. Fu, X. Gao, and L. Zhang, "Global ionospheric VTEC data completion method based on aggregated contextual-transformation generative adversarial nets," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4100523.

[17] P. Chen et al., "B3-CDG: A pseudo-sample diffusion generator for bi-temporal building binary change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 408–429, Dec. 2024.

[18] J. Feng et al., "Class-aligned and class-balancing generative domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509617.

[19] M. Chen, S. Mei, J. Fan, and M. Wang, "Opportunities and challenges of diffusion models for generative AI," *Nat. Sci. Rev.*, vol. 11, no. 12, Nov. 2024, Art. no. nwae348.

[20] X. Lu et al., "Effective variance attention-enhanced diffusion model for crop field aerial image super resolution," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 50–68, Dec. 2024.

[21] H. Cao et al., "A survey on generative diffusion models," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 7, pp. 2814–2830, Feb. 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.

[22] M. Huang, Y. Wang, Y. Wu, and Z. Jia, "Ground-penetrating radar inversion via steady-state diffusion processes," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2024, Art. no. 5101309.
[23] C. Wang and W. Sun, "Semantic guided large scale factor remote sensing image super-resolution with generative diffusion prior," *ISPRS J. Photogramm. Remote Sens.*, vol. 220, pp. 125–138, Feb. 2025.
[24] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *CoRR*, vol. abs/1409.1556, 2014.
[25] Z. Zhang et al., "Enhancing ultrahigh resolution remote sensing imagery analysis with ImageRAG: A new framework," *IEEE Geosci. Remote Sens. Mag.*, vol. 13, no. 3, pp. 369–394, Sep. 2025.
[26] C. Shorten and T. M. Khoshgoftaar, "A survey on image data augmentation for deep learning," *J. Big Data*, vol. 6, no. 1, pp. 1–48, Jul. 2019.
[27] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.
[28] M. Xu, S. Yoon, A. Fuentes, and D. S. Park, "A comprehensive survey of image augmentation techniques for deep learning," *Pattern Recognit.*, vol. 137, May 2023, Art. no. 109347.
[29] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2012, pp. 1097–1105.
[30] W. Liu et al., "SSD: Single shot multibox detector," in *Computer Vision—ECCV 2016*, B. Leibe, J. Matas, N. Sebe, and M. Welling, Eds., Cham, Switzerland: Springer, Oct. 2016, pp. 21–37.
[31] M. Cordts et al., "The cityscapes dataset for semantic urban scene understanding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3213–3223.
[32] T. DeVries and G. W. Taylor, "Improved regularization of convolutional neural networks with cutout," 2017, *arXiv:1708.04552*.
[33] S. Yun, D. Han, S. Chun, S. J. Oh, Y. Yoo, and J. Choe, "CutMix: Regularization strategy to train strong classifiers with localizable features," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6023–6032.
[34] X. Miao, Y. Zhang, J. Zhang, and X. Liang, "Hierarchical CNN classification of hyperspectral images based on 3-D attention soft augmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4217–4233, 2022.
[35] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," 2017, *arXiv:1710.09412*.
[36] M. Wang, X. Li, K. Tan, J. Mango, C. Pan, and D. Zhang, "Position-aware graph-CNN fusion network: An integrated approach combining geospatial information and graph attention network for multiclass change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4402016.
[37] Z. Xin et al., "Ship-Go: SAR ship images inpainting via instance-to-image generative diffusion models," *ISPRS J. Photogramm. Remote Sens.*, vol. 207, pp. 203–217, Jan. 2024.
[38] C. Zheng, J. Li, Y. Chen, and L. Wang, "A generalization sample learning method of deep learning for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5616515.
[39] V. Papyan, X. Y. Han, and D. L. Donoho, "Prevalence of neural collapse during the terminal phase of deep learning training," *Proc. Nat. Acad. Sci. USA*, vol. 117, no. 40, pp. 24652–24663, Oct. 2020.
[40] X.-Y. Tong et al., "Land-cover classification with high-resolution remote sensing images using transferable deep models," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111322.
[41] K. Yang, X.-Y. Tong, G.-S. Xia, W. Shen, and L. Zhang, "Hidden path selection network for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5628115.
[42] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," in *Proc. Neural Inf. Process. Syst. Track Datasets Benchmarks*, 2021.
[43] X. Zheng, L. Huan, G.-S. Xia, and J. Gong, "Parsing very high resolution urban scene images by learning deep ConvNets with edge-aware loss," *ISPRS J. Photogramm. Remote Sens.*, vol. 170, pp. 15–28, Dec. 2020.
[44] Y. Chen, L. Wang, J. Li, and C. Zheng, "Hierarchical self-learning knowledge inference based on Markov random field for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5642918.
[45] I. M. Motoi, L. Saraceni, D. Nardi, and T. A. Ciarfuglia, "Evaluating the efficacy of cut-and-paste data augmentation in semantic segmentation for satellite imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2024, pp. 9802–9806.
[46] Y. Guo et al., "SRMF: A data augmentation and multimodal fusion approach for long-tail UHR satellite image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4410916.
[47] S. Yang, H. Yang, S. Guo, F. Shen, and J. Zhao, "IPF-RDA: An information-preserving framework for robust data augmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 48, no. 2, pp. 1339–1354, Feb. 2026.
[48] C. Yu, C. Gao, J. Wang, G. Yu, C. Shen, and N. Sang, "BiSeNet v2: Bilateral network with guided aggregation for real-time semantic segmentation," *Int. J. Comput. Vis.*, vol. 129, no. 11, pp. 3051–3068, Sep. 2021.
[49] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6881–6890.
[50] R. Strudel, R. Garcia, I. Laptev, and C. Schmid, "Segmenter: Transformer for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7262–7272.
[51] W. Zhang, J. Pang, K. Chen, and C. C. Loy, "K-Net: Towards unified image segmentation," in *Proc. NIPS*, 2021, pp. 10326–10338.
[52] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, "Masked-attention mask transformer for universal image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1280–1289.
[53] H. Pan, Y. Hong, W. Sun, and Y. Jia, "Deep dual-resolution networks for real-time and accurate semantic segmentation of traffic scenes," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 3, pp. 3448–3460, Mar. 2023.
[54] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16000–16009.
[55] W. Yu et al., "MetaFormer is actually what you need for vision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10819–10829.
[56] Y. Ni, J. Liu, J. Cui, Y. Yang, and X. Wang, "Edge guidance network for semantic segmentation of high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 9382–9395, 2023.
[57] Y. Ni, J. Liu, W. Chi, X. Wang, and D. Li, "CGGLNet: Semantic segmentation network for remote sensing images based on category-guided global–local feature interaction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5615617.
[58] R. Zhang, Q. Zhang, and G. Zhang, "LSRFormer: Efficient transformer supply convolutional neural networks with global information for aerial image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610713.
[59] Y. Ni, J. Liu, H. Zhang, W. Chi, and J. Luan, "Category-guided transformer for semantic segmentation of high-resolution remote sensing images," *Remote Sens.*, vol. 17, no. 17, p. 3054, Sep. 2025.
[60] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.



Jingying Li (Graduate Student Member, IEEE) received the B.S. degree from the School of Mathematical Sciences, Dezhou University, Dezhou, China, in 2018, and the M.S. and Ph.D. degrees from the School of Mathematics and Statistics, Henan University, Kaifeng, China, in 2021 and 2025, respectively.

Her research interests include remote sensing image segmentation, deep learning, and data augmentation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.



Guoying Liu received the B.S. degree in automatic control and the M.S. degree in computer application from Changsha University of Science and Technology, Changsha, China, in 2001 and 2004, respectively, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2009.

From April 2014 to May 2015, he conducted post-doctoral research as a Visiting Scholar at the University of New Brunswick, Fredericton, NB, Canada. He is currently a Professor with the Key Laboratory OBI Information Processing of Ministry of Education, and also with the School of Software, Anyang Normal University, Anyang. His research interests include multiview clustering and oracle bone script information processing.



Jiaxuan Li received the B.S. degree from the School of Mathematics and Statistics, Henan University, Kaifeng, China, in 2022, where she is currently pursuing the M.S. degree with the School of Mathematics and Statistics.

Her research interests include deep learning and weakly supervised temporal action localization.



Tiancheng Ma received the B.S. degree from the School of Mathematics and Statistics, Henan University, Kaifeng, China, in 2022, where she is currently pursuing the Ph.D. degree with the School of Mathematics and Statistics.

Her research interests include remote sensing image segmentation, deep learning, and 3-D point cloud semantic segmentation.



Leiguang Wang received the Ph.D. degree from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan, Hubei, China, in 2009.

From 2014 to 2015, he was a Post-Doctoral Researcher at the University of New Brunswick, Fredericton, NB, Canada. He is currently a Professor with Southwest Forestry University, Kunming, China, where he has been the Vice Dean of the School of Garden and Horticulture since 2018. He has authored more than 50 articles. His research interests include remote sensing image fusion, semantic segmentation, and applications in forestry.



Chen Zheng (Member, IEEE) received the B.S. degree from the School of Mathematics and Information Sciences, Henan University, Kaifeng, Henan, China, in 2007, and the M.S. and Ph.D. degrees from the School of Mathematics and Statistics, Wuhan University, Wuhan, Hubei, China, in 2009 and 2012, respectively.

He conducted post-doctoral research at the National Astronomical Observatories, Chinese Academy of Sciences, Beijing, China, and at the University of New Brunswick, Fredericton, NB, Canada. He is currently a Professor with the School of Mathematics and Statistics, Henan University, and is in charge of a research project on semantic image segmentation funded by the National Natural Science Foundation of China. He has authored over 38 articles. His research interests include remote sensing image segmentation, data augmentation, and multilevel semantic analysis of remote sensing images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:19:43 UTC from IEEE Xplore. Restrictions apply.